MIL-F-46370A(MU)
1 April 1965
Superseding
MIL-F-46370(MU)
13 November 1962

MILITARY SPECIFICATION

FIRE CONTROL ENGINEERING SERVICES: GENERAL SPECIFICATION FOR

1. SCOPE

1.1 Scope. - This specification defines general requirements for product design studies, design of related test equipment, preparation of specifications, drawings and other technical data, and construction of engineering prototype models for Fire Control Materiel. These requirements apply in their entirety, or in part, to the extent specified in the contract or the detailed Technical Scope of Work of which this specification forms a part.

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

L-P-517	Plastic Sheet, Scribe-Coated
L-P-519	Plastic Sheet, Tracing, Glazed and Matte Finish

Military

JAN-P-100	Packaging and Packing for Overseas Shipment, General Specification For
MIL-P-116	Preservation, Methods of
MIL-D-1000	Drawings, Engineering and Associated Lists
MIL-S-11748	Interference Reduction for Electrical and Electronic Equipment
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools For Ordnance Materiel - Packaging of
MIL-F-45133	Fire Control Components & Assemblies
MIL-I-45607	Inspection Equipment, Supply and Maintenance, for Ordnance

STANDARDS

Federal

FED-STD-102	Preservation, Packaging, and Packing Levels

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Military

MIL-STD-8	Dimensioning and Tolerancing
MIL-STD-100	Engineering Drawing Practices
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-171	Finishing of Metal and Wood Surfaces
MIL-STD-194	Painting and Finishing Systems for Fire Control Instruments
MIL-STD-1174	Lists Associated with Engineering Drawings for Fire Control Materiel
MIL-STD-1175	Engineering Directives and Records For Fire Control Materiel
MIL-STD-1242	Machining Standards for Fire Control Instruments

PUBLICATIONS

Standardization Manual

M200 Chapter V	Military Outline of Form and Instructions for the Preparation of Specifications

Ordnance Pamphlet

ORD-P-20-134	Maintenance Engineering Guide to Ordnance Design

Frankford Arsenal

PP-ORD-FC-1	Provisioning Procedure for Fire Control Equipment

(Copies of specifications, standards, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

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3. REQUIREMENTS

3.1 Product engineering studies. - A thorough analysis of the existing developmental design shall be conducted in order to determine the specific areas in which redesign is necessary to achieve the following objectives:

a. Compatibility of design and detail technical data with performance requirements

b. Improved producibility and reduced unit cost

c. Maximum reliability and increased service life consistent with reasonable economic factors

d. Improved maintainability

e. Reduction in use of critical or excessively costly materials

f. Maximum use of Military Standards and existing parts

g. Elimination, to the most practicable extent, of proprietary items or sole source commercial items

h. Optimized man-machine function

3.1.1 Limitation on changes. - Except as specifically authorized by the detail Technical Scope of Work, the contractor shall make no changes in the developmental design which may adversely affect specified military characteristics or functional requirements, or which compromise in any way the reliability or maintainability characteristics inherent in the development design. To this end, the contractor shall exercise his best engineering judgment, with maximum practicable reliance on laboratory testing of breadboard models and mockups, and shall consult freely with the technical supervisor of the contract with regard to any and all product improvement proposals. The extent of product engineering is otherwise limited primarily by the mandatory schedules and by cost limitation of the contract.

3.1.2 Selection of production methods. - In selecting a specific production method, such as sand-casting, die-casting, extrusion, molding, etc., the choice shall be based upon the planned production rates specified in the detail Technical Scope of Work, as well as upon technical considerations.

3.1.3 Human engineering. - The contractor shall evaluate the developmental design for conformance with established principles of human engineering and shall recommend such changes as are considered necessary to achieve an optimized man-machine system. Consideration shall be given, but not limited to:

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a. Environmental factors such as temperature, humidity, noise, vibration, shock, dust, illumination, etc.

b. Visual and auditory perceptual requirements

c. Human safety factors

d. Operator duties, skills, educational level, and training time

e. Psychomotor requirements

3.1.4 Value engineering. - In the attainment of the objectives of improved producibility and reduced unit cost, the contractor shall conduct a part-by-part value engineering study for the purpose of ascertaining that the optimum relationship between a function and its cost has been realized. The study shall be conducted by technical personnel trained in value engineering techniques. Value engineering studies shall begin concurrently with the start of product design, and shall continue throughout the design, detailing and prototype model construction phases of the contract.

3.1.4.1 Preliminary value engineering data report. - A preliminary value engineering report shall be submitted concurrently with submission of engineering studies (Paragraph 3.1.10). This report will summarize major areas of cost reduction in the proposed redesign.

3.1.4.2 Final value engineering data report. - A final value engineering report shall be submitted within 60 days after submission of detail drawings. The report shall include a listing of each part redesigned or substituted as a result of value engineering, a brief description of its function, the estimated cost of the original development design, the estimated cost of the product-engineered version, and a brief summary of changes contributing to the cost differential. In cases where part-by-part listing is impractical, such as high-density electronic modules, the analysis may be made on a module basis rather than a component part basis. In cases of major redesign, where there is no direct substitution of one part for another, separate function-versus-cost analyses shall be made of both the development design and the product engineered version. Cost estimates shall be based on production quantities and rates specified in the detail technical scope of work. Upon approval by the Government, the report, together with any supplementary value engineering proposals accepted during the course of pilot model construction, shall be made a part of the final report required by Paragraph 3.13.2.

3.1.5 Maintenance engineering. - Ease of maintenance shall be a primary consideration in execution of the product design. Preferences and guidelines prescribed in Ordnance Pamphlet ORDP 20-134, Maintenance Engineering Guide for Ordnance Design, shall be applied to the most practical extent.

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3.1.6 Use of standard and existing parts. - Components selected for incorporation into the production design shall, insofar as possible, be standard military components and shall follow the precedence criteria of MIL-STD-143. Maximum design freedom will be permitted in the selection of components of commercial origin or in the design of components peculiar to the end item, provided that such selection or design is consistent with the requirement that maximum use be made of existing designs and parts currently in the Military system. While the introduction of additional components into the Military supply system which offer only marginal improvement over existing or standard components shall be entirely avoided, recommendations for the introduction of new items which offer marked improvement in performance, reliability, maintenance or reduced cost, are encouraged and will be favorably considered.

3.1.6.1 Approval of substitution for standard or existing parts. -

Should the contractor consider that the use of certain commercial, modified commercial, or peculiar parts, in lieu of similar standards or existing components, will offer marked improvement in performance, reliability maintenance, or reduced cost, he shall request approval of the technical supervisor prior to incorporation of such items into the final design. Requests for approval shall contain the following information:

a. Components of new design

(1) A description, or sketch

(2) A comparison of significant characteristics with those of similar existing or standard components

b. Commercial or modified commercial components

(1) A description or a reproducible print

(2) A comparison of significant characteristics with those of similar components in the Military supply system

(3) Justification for substitution

(4) Proof that the component is on the commercial market, available to the public, and is advertised in current catalogs or trade literature at an established price range.

(5) A statement that a specification control drawing will be prepared in sufficient detail to enable competitive procurement

(6) A competent opinion as to Government reprocurement rights if item is patented or otherwise proprietary

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3.1.6.2 Developmental use of non-standard parts. - The fact that certain non-standard components may have been incorporated in the original developmental design in lieu of suitable existing standard components shall not be construed as approval to incorporate these components in the production design without justification.

3.1.7 Patented or otherwise proprietary designs and processes. - Components, processes, designs, and inventions for which the Government will be subject to limited rights in data shall not be incorporated in the production design without advance disclosure of such intent by the contractor, and advance approval by the technical supervisor. Patented or proprietary designs and processes already incorporated in the development model shall be substituted to the extent specified in the Technical Scope of Work forming a part of the contract.

3.1.8 Radio interference suppression. - Design of electrical equipment shall conform with radio interference suppression requirements of MIL-I-11748.

3.1.9 Environmental requirements.

3.1.9.1 Non-operating conditions. - Unless otherwise specified in the Technical Scope of Work forming a part of the contract, equipment shall be so designed as to insure that, in the non-operating, de-energized condition, the equipment will sustain without damage or deterioration of performance, the following environmental exposure:

a. Temperature - 24 hours after thermal stabilization at 160°F and 24 hours at -80°F.

b. Humidity - 24 hours at 97 percent relative humidity and 160°F temperature followed by 4 hours at ambient temperature with condensation taking place in and on the equipment.

c. Barometric Pressure - Equivalent to atmospheric pressure at 50,000 feet above sea level.

d. Immersion or Pressure Tightness - Three feet head of water for 2 hours at ambient temperature without evidence of leakage.

e. Vibration - Simple harmonic motion at .03 inch amplitude (.06 inch total excursion) and frequency varying between 10 and 55 cps each minute, for a total of 10 minutes duration in each of three mutually perpendicular planes.

f. Shock - 30 G for 11 milliseconds duration, repeated five times.

g. Sand and Dust - 140 mesh "silica flour" at density of .3 gram $\pm$ .2 gram per cubic foot, velocity of 300 $\pm$ 200 feet per minute for six hours at temperature of approximately 77°F and relative humidity not in excess of 30%.

h. Salt Spray - 5% sodium chloride atomized spray for 48 hours at rate of approximately 2 milliliters per hour per 80 square centimeters.

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i. Fungus - Incubation period of 28 days after inoculation of all internal and external surfaces with a mixed spore suspension.

3.1.9.2 Operating conditions. - Unless otherwise specified in the Technical Scope of Work forming a part of the contract, the equipment shall be so designed as to insure capability of starting and attaining fully satisfactory operation under the following environmental conditions:

a. Temperature - -40°F and +125°F after thermal stabilization for 72 hours at each temperature.

b. Humidity - 100% relative humidity for 4 hours with condensation period of 4 hours.

c. Barometric Pressure - Ground Equipment - equivalent to 10,000 feet above sea level. Airborne Equipment - 30,000 feet above sea level.

d. Orientation - Except for equipment containing gravity operated devices, equipment shall be capable of normal operation at any orientation.

3.1.9.3 Special operating conditions. - Equipment which is so designed as to have normally closed protective covers opened or removed in the operating mode, shall be designed to operate under humidity, sand, dust and salt spray conditions specified in 3.1.9.1. In addition, the equipment shall be capable of satisfactory operation when exposed to rain fall of 4 to 5 inches per hour for 2 hours with a wind velocity of 20 mph directed at surfaces normally exposed to rain during operation of the equipment.

3.1.10 Submittal of engineering studies. - Engineering studies shall be submitted by the contractor for approval prior to preparation of detail drawings. Unless otherwise specified in the Technical Scope of Work, engineering studies shall be submitted in triplicate, including one reproducible copy, and shall consist of:

a. An analysis of the development design to identify key factors affecting producibility, reliability, function and maintainability.

b. A summary of changes proposed in the production design, and the basis for the proposed changes.

c. Schematic diagrams showing basic concept and circuit changes.

d. Design layouts drawn to scale in third angle orthographic projection and in sufficient detail to serve their primary purpose of providing guidance to detail draftsmen. They should clearly show space allocations, and areas of interface with other assemblies or other major weapon system components.

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e. An analysis of major functional requirements, critical strength and environmental considerations, and computation of weight

f. A preliminary value engineering report containing a detailed cost comparison between the developmental model and the proposed redesign

3.1.10.1 Partial submission. - Engineering studies may be submitted as a complete study or as a portion of the overall project, if breakout of any portion of the project is practical. All breadboards and mock-ups considered necessary to demonstrate feasibility shall be made available to the technical supervisor at the time of submission of the design study.

3.2 Engineering drawings. - The contractor shall, upon approval of the engineering study, prepare a complete set of engineering drawings in conformance with the requirements of Form 1, MIL-D-1000. Drawings shall be prepared in pencil on tracing cloth or pencil on matte finish plastic sheet. Engineering drawings conforming with the requirement of Form 2, MIL-D-1000, will be accepted only if such drawings are shown to have been in existence prior to the issuance of the invitation for bid or request for proposal. Form 3 drawings will not be acceptable under this specification unless otherwise permitted by the contract.

3.2.1 Intended use. - Engineering drawings prepared under this specification shall be suitable for the following categories of intended use, as defined in MIL-D-1000:

3.2.1.1 Category E - Procurement of identical items. - Drawings suitable for procurement of identical items shall be prepared for:

a. Complete end items

b. All sub-assemblies

c. All components except components of sub-assemblies covered by authorized "specification control" and "source control" drawings

d. All repair parts for maintainable sub-assemblies covered by "specification control" and "source control" drawings

3.2.1.2 Category G - Installation - Installation drawings shall be prepared for each fire control major item or combination of items which are intended to be installed in a vehicle, weapon, or facility.

3.2.1.3 Category H - Maintenance. - Drawings in this category shall be prepared for each fire control major item when drawings which are fully adequate for procurement do not provide sufficient information for effective maintenance.

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3.2.2 Selection of types of engineering drawings. - Selection of the types of engineering drawings shall be governed by MIL-STD-100 and the following supplementary requirements:

3.2.2.1 Detail drawings. - Except as otherwise provided below, a mono-detail drawing shall be prepared for each identifiable part.

3.2.2.2 Inseparable assembly drawings. - An inseparable assembly drawing may be prepared in lieu of individual mono-detail drawings for inseparable (welded, brazed, bonded, riveted, sewn, or glued) assemblies intended to be procured and replaced as a unit, where (except for standard hardware) the separate parts are of the same material and where each part is manufacturable by a similar process. Example: A welded or riveted bracket entirely constructed of the same material, a wooden, metal, or plastic chest, or a canvas case may be covered by an inseparable assembly drawing without mono-detail drawings. A glass prism bonded to an aluminum plate would require support by mono-detail drawings. Mono-detail drawings shall be used whenever they are necessary for clarity of complex configurations.

3.2.2.3 Tabulated drawings. - Tabulated drawings may be prepared in lieu of individual detail drawings as appropriate, provided that clarity is not adversely affected.

3.2.2.4 Specification control drawings. -

a. Specification Control Drawings shall be the preferred method of delineating vendor-developed items provided that such items are advertised or catalogued as available to the trade or to the public on an unrestricted basis at an established range of price. In addition, they may be used for vendor-developed items which, while not commercially available, are procurable on order from two or more suppliers in a specialized segment of the industry. Typical examples of items in this category are special motors, synchros, transformers, potentiometers, hydraulic valves, carburetors, potted servo amplifiers, keyboards, tape readers and similar products of industries which normally provide customer applications engineering services for a commercial product line.

b. Unless otherwise provided in the contract or specifically approved by the technical supervisor, Specification Control Drawings will not be acceptable in lieu of detail disclosure for items which will normally require replacement of any of its component parts during the life of the materiel. Any exception to this requirement will require that, as a minimum, mono-detail drawings be supplied by the contractor for all spare parts.

c. Notwithstanding the provisions of this paragraph, the Government reserves the right to require fully detailed design disclosure for any component developed under the contract.

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d. Specification Control Drawings shall not be prepared under circumstances where a Military Detail Specification would be more appropriate. (3.4)

e. Specification control data shall not be tailored to the characteristics of a single vendor's product to the exclusion of other equally suitable products. Conversely, specification data shall not be so broad as to permit acceptance of products which will not perform in the equipment under all required environmental conditions.

3.2.2.5 Source control drawings. - Source control drawings shall be prepared only upon written authorization of the technical supervisor. Requests for authorization to use Source Control Drawings shall be accompanied by a written justification setting forth the technological or other factors beyond the control of the contractor which prevent the determination of performance, reliability and configuration control requirements needed to prepare a Specification Control Drawing. This justification will form a part of the permanent engineering file.

3.2.2.6 Undimensioned master drawings. - Undimensioned master drawings shall be supplied for etched circuit boards, panels, reticles, special scales, and other components for which dimensioned detail drawings would be impractical. Master drawings shall be prepared on dimensionally stable plastic sheet meeting the requirements of L-P-517 or L-P-519. Drawings shall be of high quality, clearly drawn, typed, or scribed with sharp, clean-cut lines to insure a high degree of legible reproducibility. Appliqued patterns shall be photographically opaque except for desired open areas. Unless otherwise specified in the contract, the contractor shall supply plastic sheet for master drawings in accordance with standard format requirements.

3.2.2.7 Matched parts drawings. - Matched parts drawings shall be prepared as required and will normally be supported by mono-detail drawings. Each pertinent mono-detail drawing shall clearly indicate its dependence upon the matched parts drawing.

3.2.2.8 Envelope drawings. - Envelope drawings shall be prepared only to support a Military Detail Specification. In all other cases, envelope drawings shall be prepared as Specification Control or Source Control drawings.

3.2.2.9 Assembly drawings. - Assembly drawings shall be prepared for the major end item and for each level of sub-assembly required for spare parts supply, bench sub-assembly and testing, localization of major functional requirements and adjustments, and clarity of parts identification, assembly and diagnostic procedures. Insofar as practical, each sub-assembly drawing shall include pertinent performance specifications.

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3.2.2.10 Electrical diagrams. - A schematic drawing shall be prepared for each electrically functional sub-assembly; i.e. one having a measurable input and output relationship. A connection or wiring diagram, or, for more complex electronic items, a wiring list, shall be prepared for each sub-assembly stage covered by an assembly drawing.

3.2.2.11 Mechanical schematic diagram. - A mechanical schematic diagram shall be prepared for each major item or functional assembly comprising a complex arrangement of gears, clutches, linkages, cams, etc. where operating principles cannot readily be derived from a study of assembly drawings.

3.2.2.12 Installation drawing. - See 3.2.1.2

3.2.3 Dimensions and tolerances.

a. Dimensions and tolerances shall be specified in accordance with MIL-STD-8 and shall be selected to assure that build-up of maximum tolerances will not cause functional or assembly difficulties. Where fitting at assembly or selective assembly is intended, the detail drawing of each component shall clearly indicate the specific dimension or dimensions which will require fitting or selection at assembly, and shall indicate the fitting or selection criteria.

b. Tolerances shall be as liberal as possible consistent with assembly, functional and reliability requirements, but under no circumstances shall the desirability of maximizing tolerance limits on a component be permitted to compromise the feasibility of assembly and adjustment, or to compromise performance requirements of the assembled item. Conversely, compliance with assembly and functional requirements shall not be "legislated" by specifying tolerances which are impractical to attain by specified or implied production processes. Such incompatibilities shall be corrected by a change in design or performance requirements.

c. Tolerances for flatness, perpendicularity, parallelism, concentricity and roundness of machined surfaces may be omitted if standards established by MIL-STD-1242 are suitable, provided that MIL-STD-1242 is referenced in the drawing or in an applicable specification. Inclusion of Specification MIL-F-13926 or MIL-F-45133 in the applicable list of specifications will automatically apply the provisions of MIL-STD-1242.

d. Use of the basic hole system of tolerancing (MIL-STD-8) is preferred over the basic shaft system.

3.2.4 Protective finishes.

3.2.4.1 Selection. - Protective organic and inorganic finishes for metal and wood surfaces shall be selected from MIL-STD-171 and supplementary limited coordination military standard MIL-STD-194[illegible]

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3.2.4.2 Dimensioning of finished surfaces.

a. Surfaces protected by inorganic chemical finishes (plating, anodize, oxide, phosphate, etc.) shall be dimensioned to include any build-up resulting from the finish process. It will be the manufacturer's responsibility to allow for finish build-up in machining of the surfaces.

b. Surfaces protected by organic or other paint finishes shall be dimensioned to exclude the build-up resulting from painting.

3.2.5 Drawing formats. - Except as provided in 3.2.2.6, preprinted drawing formats for use in the preparation of Form 1 drawings will be supplied by the Government upon application by the contractor accompanied by an estimate of the number of each size of drawing format which will be required.

3.2.6 Drawing numbers. - A block of drawing numbers will be provided by Frankford Arsenal upon request of the contractor. Appropriate request forms will be supplied by the technical supervisor. Drawing number cards issued by the Government shall be carefully safeguarded and controlled by the contractor against loss. The contractor shall enter the required nomenclature and application data on each card and shall return all used and unused drawing number cards to Frankford Arsenal upon completion of the contract or at such earlier time as may be specified by the technical supervisor.

3.2.7 Code identification. - The code 19200 shall be entered on all Form 1 drawings as required by MIL-STD-100.

3.3 Supplementary quality assurance provisions (SQAP). - A SQAP sheet shall be prepared for each part falling within the following criteria, unless this specification or the Technical Scope of Work forming a part of the contract requires the preparation of a Military Specification (3.4) or unless a Military Specification already exists:

a. Components or assemblies designated as spare parts except when inspection of characteristics requires only the use of standard measuring equipment.

b. Components which mate with a spare part except when inspection of characteristics requires only the use of standard measuring equipment.

c. Components which have interfaces with other components of a weapon system.

d. Components in which discrepancies or variations from drawing requirements might result in a critical safety hazard.

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e. Components in which discrepancies or variations from drawing requirements would not be detectable in normal functional inspection of the end item, but which could result in performance or reliability deficiencies in the field.

f. Components in which quality is dependent upon rigid adherence to specified physical or chemical properties of raw or finished materials.

g. Components for which it is necessary or desirable to prescribe specific test methods and procedures, or for which the use of specific test equipment designs is mandatory.

3.3.1 Format and content of SQAP. - SQAP sheets shall be prepared in accordance with the requirements of Appendix III of this specification.

3.4 Military specifications. (See also 3.5.4.2) - Military detail specifications shall conform with the requirements of Chapter V, Standardization Manual M200, and this specification. A specification shall be prepared for:

a. The major item(s) and system described in the Technical Scope of Work.

b. Components and intermediate assemblies which can be expected to be repetitively procured as spare parts and for which the necessary performance requirements and quality assurance provisions are lengthy and complex. As a general guide, it shall be assumed that any spare part requiring performance specifications in excess of two single-spaced typewritten pages, or supplementary quality assurance provisions in excess of four typed pages will be covered by a Military Specification in lieu of drawing notes or a SQAP.

c. Components and intermediate assemblies for which a Qualified Products List (QPL) will be required.

3.4.1 Form of specifications. - Fire Control specifications prepared under the requirements of this specification shall be "Detail Specifications" as defined in Chapter V, Standardization Manual M200, and shall be one of the following two types of detail specifications.

3.4.1.1 Detail design specification. - Unless otherwise specified in the Technical Scope of Work of the contract or as otherwise approved by the technical supervisor, specifications shall be of the "detail design" type; i.e. in addition to performance requirements and quality assurance provisions controlling end item performance, the specification shall include by mandatory reference, a complete set of engineering drawings as specified in 3.2.

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3.4.1.2 Detail performance specifications. - When authorized by the technical supervisor, and when preparation of a Performance Specification is required under criteria specified in 3.4, certain components and intermediate assemblies may be treated as "black-boxes" and covered by performance and envelope requirements only. Circumstances under which Detail Performance Specifications are permitted or preferred in lieu of Detail Design Specifications are identical to those governing the preparation of Specification Control Drawings in lieu of detail drawings. (3.2.2.4)

3.4.2 Specifying requirements (Section 3). - Instructions for preparation of Section 3 of Fire Control Military Specifications are contained in Appendix I of this specification.

3.4.3 Quality assurance provisions. - Quality Assurance Provisions, Section 4, of Fire Control specifications shall be prepared by the contractor in accordance with Appendix II of this specification.

3.4.4 Preliminary draft. - A preliminary draft of each required specification shall be submitted to Frankford Arsenal for review and approval. The draft shall be typed, double space on 8" x 10½" white bond paper and two copies. Unless otherwise specified in the contract, drafts shall be submitted no later than 60 days after approval of engineering studies.

3.4.5 Final copy. - The final copy of each required specification will be prepared by the contractor based on comment of Frankford Arsenal. Copy shall be typewritten, on 8 x 10½" bond paper. Illustrations, if any, will be integrated into the body of the copy as required. Copy will be suitable for photographing for lithographic reproduction.

3.4.5.1 Typing. - Unless otherwise specified, copy shall be typed using elite type, 12 characters per inch. Typing shall be sharp, clean and of uniformly high density. Copy shall be free of over-strikes and typographical errors.

3.4.5.2 Illustrations. - Illustrations used in specifications shall be of the contrast and clarity required for satisfactory reproduction by the lithographic process, and shall be mounted flat without bulges or ripples.

3.5 Inspection equipment. - The contractor shall design, and prepare complete technical documentation for special test equipment required for inspection of the end item and selected components or intermediate assemblies. Unless otherwise specified in the detail Technical Scope of Work, design and documentation will be limited to Class A and B test equipment, as defined in 3.5.1, except that existing Class C and E test equipment shall be included in pertinent specifications and data lists.

3.5.1 Classification of inspection equipment.

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3.5.1.1 Class A test equipment. - This class is applicable to functional test equipment which is essential to the completeness and adequacy of quality assurance specifications, i.e. where performance of the product can be practically measured only in terms of its performance under specific conditions embodied in the test equipment; where a verbal or schematic description of the equipment in Section 4 of the specification or SQAP would be difficult to prepare and to understand, excessive in length, or require a high degree of technical training to interpret without the aid of a detail design.

3.5.1.2 Class B test equipment. - This class is applicable to functional and dimensional test equipment required for the inspection of components or assemblies having characteristics classified as "critical", whether or not test methods and procedures can be otherwise specified in Section 4 of the specification, or in SQAP. This class also includes interchangeability gages for checking mounting interfaces between a major item and the equipment or weapon in which it is installed or with which it mates.

3.5.1.3 Class C test equipment. - This class is applicable to functional test equipment for which basic constructional requirements are verbally or schematically specified in Section 4 of the major item specification or in component SQAP, in accordance with preferred practice, and for which the recurring cost of design and documentation on each future production contract would be $10,000 or higher, if technical documentation were not available.

3.5.1.4 Class D test equipment. - This class is applicable to test equipment for which constructional and functional requirements are verbally or schematically specified in Section 4 of the major item specification or in component SQAP, in accordance with preferred practice, and for which the recurring cost of design and documentation on each future production contract would be less than $10,000 if technical documentation were not available. This class is also applicable to all dimensional and geometric gages for non-critical characteristics. Unless otherwise specified in the detail technical scope of work, it shall be assumed that the design and supply of Class D test equipment will be the responsibility of the future production supplier, in accordance with the verbal or schematic description contained in specifications and SQAP.

3.5.1.5 Class E test equipment. - This class is applicable to test equipment which would normally be classified Class D, but for which technical documentation has been, or is being, developed and maintained for Field Maintenance purposes. (3.7)

3.5.2 Design requirements.

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3.5.2.1 Simplicity and reliability. - Test equipment shall be of maximum simplicity, dependability and reliability, and shall be designed to minimize requirements for maintenance, calibration, and special operator skills.

3.5.2.2 Accuracy. - To the extent practical, test equipment shall be designed to be accurate to within 10% of the product tolerance. Concept, construction, materials, dimensions, and tolerances used in design of test equipment will permit positive rejection of a product which exceeds the prescribed tolerance limit and will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit; i.e. the gage tolerance will be taken in the direction of rejection.

3.5.2.3 Calibration. - Construction shall be such as to facilitate routine calibration of the equipment.

3.5.3 Submittal of test equipment designs.

3.5.3.1 Analysis of test equipment requirements. - Not later than 30 days after submittal of product engineering studies, the contractor shall prepare an analysis of acceptance test equipment requirements. The analysis shall include, a brief description of the proposed equipment, accuracy requirements, estimated cost of design and documentation, and estimated cost of construction of one prototype of each.

3.5.3.2 Design studies. - Upon approval by the Government of the analysis of test equipment requirements, the contractor shall proceed with detailed design analysis and layout drawings. The complete design study for any single item of test equipment shall be submitted for approval prior to preparation of detail drawings. Unless otherwise specified in the Technical Scope of Work, the design studies shall consist of:

a. Schematic diagrams showing basic concept, optical system, circuitry, etc.

b. An analysis of the theory of operation.

c. Design layouts drawn to scale in third angle orthographic projection, and in sufficient detail to serve their primary purpose of providing guidance to detail draftsmen.

d. Set-up and operating instructions

3.5.4 Test equipment documentation.

3.5.4.1 Drawings. - Upon receipt of approval of design studies, the contractor shall prepare a complete set of engineering drawings in accordance with 3.2, except as follows:

a. Multi-detail drawings shall be prepared in lieu of mono-detail drawings; dash numbers shall be used to identify separable details.

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b. Drawings may be prepared in pencil on translucent bond paper in lieu of linen or plastic sheet.

c. Detail-assembly drawings may be used provided clarity is not adversely affected.

d. Preparation of sub-assembly drawings shall be minimized consistent with clarity.

e. Source control drawings may be prepared where the cost of developing specification control data is not economically justifiable.

f. A separate block of test equipment drawing numbers will be provided by Frankford Arsenal upon request of the contractor. (3.2.6)

3.5.4.2 Specifications. - Insofar as practicable, detail performance requirements and inspection requirements for test equipment will be included on the major assembly drawing. Where these requirements are lengthy and complex, a military detail specification shall be prepared by the contractor in lieu of drawing notes. As a general guide, a specification shall be prepared whenever the functional and inspection requirements would exceed four typewritten letter-size pages. Set-up and operating instructions for use of the equipment shall not be included in the military specification. When required, specifications shall be prepared in accordance with requirements of 3.4.

3.6 Engineering lists. - The contractor shall prepare the following engineering lists in accordance with the requirements of MIL-STD-1174, using blank forms supplied by the Contracting Officer.

3.6.1 Engineering parts list. - An Engineering Parts List shall be prepared for each major item, for each intermediate assembly designated a spare part, and for each item of inspection or Field Test Equipment. The Engineering Parts List shall consist of:

a. EPL - Contents

b. EPL - List of Parts

c. EPL - Listing of Parts Lists

d. EPL - List of Drawings (Cumulative)

e. EPL - List of Specifications (Cumulative)

f. IEL - List of Inspection Documents (Cumulative)

3.6.2 Sub-assembly engineering parts list. - An EPL - List of Parts shall be prepared for each first level and intermediate sub-assembly.

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3.6.3 EPL - List of specifications for components. - An EPL list of specifications shall be prepared for each drawing covering a component designated as a spare part. This list shall include all specifications necessary for the procurement and inspection of the component.

3.7 Field and depot test equipment and tools. - The contractor shall design and prepare technical documentation for all Field and Depot test equipment, special tools and fixtures specifically required by the detail Technical Scope of Work.

3.7.1 Design. - Insofar as practical, the design of Field and Depot test equipment and the design of Inspection Equipment (See 3.5) shall be identical.

3.7.2 Design studies. - Design studies shall be submitted by the contractor for each item of field or depot test equipment. Unless otherwise specified in the Technical Scope of Work, the design studies shall consist of:

a. Schematic diagrams showing basic concept, optical system, circuitry, etc.

b. An analysis of the theory of operation.

c. Design layouts drawn to scale in third angle orthographic projection, and in sufficient detail to serve the primary purpose of providing guidance to detail draftsmen.

d. Set-up and operating instructions.

3.7.3 Documentation. - Drawings and specifications shall be prepared by the contractor in accordance with 3.5.4.

3.7.4 Engineering lists. - Engineering lists shall be prepared for each item of test equipment in accordance with 3.6.1, to the extent applicable.

3.8 Packaging data. - Packaging, packing, and marking requirements shall be developed for each major item, each item of test equipment, and each component designated as a repair part except for items where the packaging requirements are presently prescribed in a specification.

3.8.1 Design. - Packaging shall be designed for Levels A and C and packing shall be designed for Levels A, B and C in accordance with FED-STD-102, MIL-P-116 and MIL-P-14232. When reusable containers are required by the detail Technical Scope of Work, the complete design study for any single container shall be submitted for approval prior to preparation of detail drawings.

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3.8.2 Format. - Packaging instructions shall be prepared on packaging data sheets AMC Forms 1029 and 1029-1, in accordance with Appendix III, for use with MIL-P-14232. Preprinted packaging data sheet forms will be supplied by the Government, upon application by the contractor accompanied by an estimate of the number of packaging data sheets which will be required.

3.8.3 Where detailed engineering drawings are required to properly prescribe packaging requirements, in lieu of packaging data continuation sheets, they shall be prepared in accordance with 3.2.

3.9 Data approval and release.

3.9.1 Release documentation. - The contractor shall prepare an "Initial Release" Engineering Order (EO) and (EO) Inclosure List for each "major item" of test equipment in conformance with the requirements of MIL-STD-1175. Engineering Orders and Engineering Notices shall be typed on standard formats using carbon backing, or other suitable method, to obtain maximum density required for reproduction by normal contact printing methods. Standard forms and Engineering Order serial numbers will be issued by the contracting officer upon request.

3.9.2 Approval. - At such time as the contractor shall consider all or part of the drawings and other documentation to be ready for approval, the originals, together with two reproducible prints of each document shall be hand-carried or otherwise reliably delivered to Frankford Arsenal for approval signature and official release. Unless otherwise specified in the detail technical scope of work or by the technical supervisor, the signed original drawings will be returned to the contractor for maintenance for the duration of the contract.

3.9.3 Microfilm aperture cards. - If specifically required in the contract, the contractor shall supply six punch-card mounted camera original microfilm reproductions of each drawing, packaging data sheet, engineering list, and SQAP, in accordance with the requirements specified in the Technical Scope of Work. Such microfilm cards shall be furnished by the contractor within a maximum of 21 calendar days after receipt of signed original engineering documents.

3.10 Prototype manufacture. - The following requirements are applicable where the detail technical scope of work or the contract specifically requires the manufacture of one or more prototypes of the major item, its protective package, inspection equipment, or Field Service tools and test equipment.

3.10.1 Authorization. - Government approval of pertinent drawings and specifications shall constitute authorization to proceed with manufacture or purchase of component parts. Manufacture or purchase of components for which technical data has not yet been approved will be permitted only by written authorization of the Contracting Officer.

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3.10.2 Use of approved engineering documentation. - In order to test the adequacy of drawings and specifications for manufacturing purposes, and to assure fabrication in conformance with the intent of the design, manufacture or purchase of components for prototypes shall be on the basis of Government approved drawings and specifications, including those which are referenced within these drawings and specifications. No other technical data, Industry or company specifications or standards, or oral technical instructions not intended to be delivered or disclosed to the Government, shall be used to supplement the drawings or specifications during manufacture, purchase, or inspection. This restriction shall not apply to use of administrative procedural specifications or standards normally used by the contractor in manufacture or purchase of components; nor is this restriction intended to apply to manufacturing details pertaining to "Specification Control" drawings or "Source Control" drawings which have been approved for use.

3.10.3 Performance requirements. - The prototype model together with its packaging and any required test equipment shall conform with the detail design and performance requirements specified in the Technical Scope of Work. Unless otherwise specified, all functional and reliability characteristics of the product engineered prototype shall be equal to or better than those of the developmental model, as evidenced by Engineering Test and User Test reports, development specifications, and independent evaluation by the contractor.

3.10.4 Detail requirements. - Manufactured and purchased components shall conform with dimensions, tolerances, materials, finish, and functional requirements specified in approved technical data. No variations between components and approved technical data will be permitted unless authorized by a Request for Waiver or an engineering order in accordance with 3.11.1.

3.11 Maintenance of technical data. - Unless otherwise specified in the detail Technical Scope of Work, or until physical custody of the original documentation is assumed by Frankford Arsenal, the contractor shall maintain all drawings, specifications, SQAP, packaging data sheets, and engineering lists in an up-to-date status for the duration of the contract.

3.11.1 Minor revisions during prototype phase. - The contractor shall expeditiously revise technical data to include such changes as may be found necessary during manufacture, inspection, and special testing of the prototype model. He may, without prior approval, revise approved Government drawings in his custody, for minor changes in design, dimensioning, tolerances, materials, correction of drafting errors, etc. In such cases the contractor shall prepare and submit a complete Engineering Order Package consisting of one reproducible print of the revised drawing(s), a fully descriptive Engineering Order, Engineering Notices, and inclosure list in accordance with the requirements of MIL-STD-1175. In the event of non-concurrence by the Government in the revision, the drawings shall be

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returned to their previous state or otherwise revised in accordance with instructions of the technical supervisor. The approved Engineering Order or copy thereof will serve as authorization to the Government inspector to accept prototype material conforming with the revised drawing or specification.

3.11.2 Major revisions during prototype phase. - The contractor shall make no changes to any approved drawing or specification which may adversely affect function, service life, maintainability, or human factors, or which will increase the cost of the prototypes by more than 1%. Proposed changes in these categories will be submitted for the consideration of the technical supervisor by means of an Engineering Change Request (ECR) in accordance with MIL-STD-1175. Technical data shall be revised only upon receipt of a copy of the approved ECR.

3.11.3 Revisions during production phase. - The following requirements are applicable only where the contractor is specifically required to maintain custody of technical data subsequent to release for quantity production.

3.11.3.1 Engineering change requests. - Drawings or other technical data forming a part of a Technical Data Package which has been released for quantity production may be revised only on authority of a Government-approved Engineering Change Request. Such requests shall be expeditiously prepared and submitted by the contractor in accordance with MIL-STD-1175 when, as a result of production experience or test results, changes are found to be essential or desirable. In addition to ECRs originated by the contractor, the Government may originate or request the preparation of ECRs. The contractor shall also make changes to technical data, and prepare a complete EO package, on the basis of Government approval of Technical Action Requests (Request for Technical Action, AMC Form 1020-1), originating from any supply contractor. A copy of such approved Requests for Technical Action shall constitute an authorized Engineering Change Request.

3.11.3.1.1 Changes affecting current production. - Engineering Change Requests shall not be prepared for application to current production unless they are unquestionably essential, or will result in a significant reduction in contract price. An essential change will be defined as one which serves to correct:

a. A condition which will lead to malfunction of the item

b. A condition which will result in failure to attain required reliability

c. A condition which may create a safety hazard to personnel during operation, installation, storage, or transportation

d. An impossible manufacturing or assembly condition

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e. An incompatibility between design and performance requirements

f. A serious logistics or maintenance problem

An Engineering Change Request intended solely to reduce production costs may be classified for application to current production only if it will unquestionably result in a reduction in contract price which is significantly greater than the cost of introducing the change into production.

3.11.3.1.2 Changes to be applied to future production. - Engineering changes which do not meet the criteria specified in 3.11.3.1.1, if considered desirable and justifiable from the point of view of improvement in performance, producibility, and maintainability shall be classified for "future contracts" application only.

3.11.3.2 Engineering orders. - Engineering Orders and revisions to drawings and specifications shall be processed promptly upon receipt of an approved ECR. Engineering Orders classified for release to existing contracts shall be processed within a maximum of 10 days unless otherwise specified by the technical supervisor. Engineering Orders classified for future contracts shall be processed within a maximum of 30 days. Engineering Orders shall be prepared in accordance with the requirements of MIL-STD-1175[illegible].

3.12 Provisioning list (spare parts). - The contractor shall prepare a tentative provisioning list containing those assemblies and components which in his best engineering judgment, will be required as spare parts. The provisioning list shall conform with the requirements of Frankford Arsenal Publication PP-ORD-FC-1, Provisioning Procedures for Fire Control Equipment. Unless otherwise specified in the contract, the first draft of the provisioning list shall be submitted no later than 30 days after design approval, and shall be maintained in an up-to-date status during the detailing phase and subsequently, as necessary.

3.13 Reports.

3.13.1 Monthly progress reports. - A narrative progress report shall be prepared monthly during the life of the contract. When required by the contract, four copies, including one reproducible copy, shall be forwarded to the Commanding Officer, Frankford Arsenal, Philadelphia, Pa. 19137, addressed to the attention of the designated technical supervisor. While no standard format is required, the report shall include, as a minimum a description of work performed, technical difficulties encountered or anticipated, progress against detailed plan of accomplishment, forecast of accomplishment for the succeeding month, areas of present or foreseeable schedule slippage and planned corrective action, expenditures incurred during the last reporting period and cumulative expenditures to date. The first report shall include a chart showing the detail plan of accomplishment of each important activity contributing to the successful and timely completion of the contract. In each subsequent report, the chart shall show progress made against each planned activity. Any necessary revision of the plan shall be explained in the narrative report.

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3.13.2 Final report. - A final report shall be submitted by the contractor at least 30 days in advance of planned completion of the contract or at such earlier date as may be designated by the technical supervisor.

3.13.2.1 Content. - The report shall include highlights of each of the monthly reports, rearranged and edited for logical continuity and grouping of subject matter. It shall include a summary of all work performed; a detail analysis of the basis for selection of final design approach; and substantiating data in the form of notes, calculations and sketches pertaining to salient design features. It shall include the final value engineering report. (3.1.4.2)

3.13.2.2 Format. - Although no specific format will be required, the report shall begin with an extract or summary edited and illustrated specifically for top management and administrative personnel with the primary purpose of demonstrating the benefits gained from the engineering funds expended. The use of appendices are recommended for recording of calculations, analyses, test results, etc., to insure maximum readability of the body of the report. The report shall be carefully prepared, edited and assembled to serve not only as evidence of worthy project achievement, but as a permanent record for future reference in making engineering decisions relating to this project or related projects.

3.13.2.3 Reproduction. - Unless otherwise specified, the report shall be typewritten on 8" x 10½" bond paper, stapled or otherwise permanently bound between paper covers. Reports may be bound in more than one volume, if desired. Six copies of the complete report and twelve additional bound copies of the extract or summary shall be submitted by the contractor to the technical supervisor. Reports shall be reproduced by Xerographic (electrostatic carbon) or by paper lithographic process. One of the six copies shall be reproduced on translucent paper and shall be suitable for reproduction by normal blueprint or "Ozalid" contract printing processes.

3.14 Liaison requirements.

3.14.1 Liaison with the technical supervisor. - The contractor shall maintain adequate liaison with the technical supervising agency to insure efficient interchange of information and prompt transmittal of data submitted for approval, including the hand-carrying of data when required. Unless otherwise specified in the Technical Scope of Work, or unless otherwise directed by the Technical Supervisor, qualified personnel of the contractor will visit the technical supervising agency at least once each month during the life of the contract for the purpose of technical consultation and review of problems and progress.

3.14.2 Technical assistance by the contractor. - The contractor shall, upon the written request of the technical supervisor, visit such industrial facilities, military installations, or other Government agencies as may be designated, for the purpose of participating in tests and demonstrations, providing field repair and maintenance services, or otherwise acting on

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behalf of, or in support of the technical supervisor, on tasks directly related to the technical scope of the contract.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 General provisions.

4.1.1.1 Technical data. - The contractor shall provide and maintain an adequate system for the detailed checking and technical review of all design studies, drawings, SQAP, engineering lists, specifications, and packaging data to be supplied under the terms of the contract. A written description of the checking and review system to be used shall be submitted to the Contracting Officer for approval within 30 days after award of the contract. Changes in the system shall thereafter be made only with the approval of the Contracting Officer. The minimum requirements to be included in the contractor's written checking system shall be as follows:

a. Organizational assignment of responsibility for detail checking, review and final contractor approval of technical documentation.

b. Resumes of qualifications of design checkers, detail drawing checkers, specification reviewers, packaging specialists, quality engineering specialists, and personnel having final approval authority.

c. A description of the procedural flow of all technical documentation including the extent and depth of review at each echelon of concurrence and approval, and including:

(1) Material selection review procedures

(2) Value engineering review procedures

(3) Standardization review procedures (use standard and existing parts)

d. Procedures to be used for "worst case" analysis of design.

e. A copy or description of check-lists and guides to be used in the detail checking, review, and approval of technical documentation.

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f. Identification of all characteristics which could result in Critical Defects, and a resume of the qualifications of the professional engineer(s) who will attest to the accuracy of critical data.

g. Procedures to be used in sampling inspection of technical data prior to submission to the Government, and a resume of qualifications of personnel who will conduct the inspection.

4.1.1.2 Prototype models. - The contractor shall provide and maintain an adequate inspection system, acceptable to the Contracting Officer, covering the inspection of components and final testing of the prototype model. The contractor shall perform all inspections required by this inspection system. A written Quality Assurance Plan shall be submitted to the Contracting Officer prior to initiation of prototype manufacture. The written plan will be considered acceptable when it provides as a minimum, the assurance that prototypes, and all components thereof, are inspected and tested in strict conformance with approved technical data and authorized changes thereto. Changes in the plan will thereafter be made only with the approval of the Contracting Officer. The minimum requirements to be included in the contractor's Quality Assurance Plan shall be as follows:

a. An outline summary of the inspection organization, qualifications of inspectors, and management officials to whom they report.

b. Method of recording inspection results

c. Procedures for segregation and disposition of non-conforming material.

d. Control of engineering changes and requests for waiver.

4.2 Inspection provisions for technical data.

4.2.1 Checking.- The contractor shall conduct a 100% check of all drawings, lists, SQAP, and specifications prior to submission to the Government for acceptance. Checking shall be conducted by authorized personnel (Paragraph 4.1.1.1) other than the person or persons who prepared the document. The checker's initials on the original drawing, or signature on the cover sheet of a specification, shall constitute a certification of compliance with all requirements of the contractor's written checking system, and with all requirements of this specification and the Technical Scope of Work forming a part of the contract.

4.2.1.1 Certification of critical data. - The accuracy of technical data in which critical defects (Paragraph 4.2.2.4.1)may occur shall be certified by a qualified engineer of recognized standing in the profession.

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4.2.2 Examination subsequent to 100% check. - Technical documentation will be accepted by the Government on the basis of examination of a sampling of each lot of drawings submitted by the contractor. The criteria specified in Table I shall constitute the minimum independent re-checking by the contractor subsequent to the 100% check required by 4.2.1, and will constitute the maximum severity of examination by the Government for acceptance or rejection by lot. The Government reserves the right to conduct 100% inspection of a lot or of any portion of a lot, and to reject all individual non-conforming documents. Technical data in which Critical defects may occur will be subject to 100% examination by the Government.

TABLE I
LOT ACCEPTANCE CRITERIA

LOT SIZE	SAMPLE	MAJOR DEFECTS PERMITTED	MINOR DEFECTS PERMITTED
1 - 5	100% Inspection	-	-
6 - 25	5	0	1
25 - 40	7	0	1
41 - 65	10	0	2
66 - 100	15	0	3
101 - 200	25	1	4
201 - 300	35	2	7
301 - 500	50	3	9
501 - 800	75	4	13
801 - 1300	110	6	18

4.2.2.1 Lot formation. - For purposes of acceptance or rejection by lot, a lot shall consist of any number of related documents prepared and checked under common supervision in a single organization governed by a single set of procedures, policies, and standards. Engineering drawings, engineering lists, SQAP, and specifications shall be submitted as separate lots for examination purposes. The Government reserves the right to combine submitted lots to form a single lot, or to divide a single lot into smaller lots, as may be required to facilitate sampling inspection. For purposes of lot formation, a single document constitutes all sheets and pages comprising that document.

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4.2.2.2 Sampling of engineering drawings, engineering lists, and SQAP. - Each lot will be arranged in numerical sequence. The frequency of sampling will be determined by the ratio of the size of the lot to the size of the sample as specified in Table I. For example, if the lot size is 150, the sample size will be 25, and the ratio of lot size to sample size will be 6; therefore, every sixth document will be selected for inspection. The first sample will be selected at random from the first six documents in numerical sequence. If the ratio between lot size and sample size were 9, one of the first 9 documents would be selected at random followed by every ninth in numerical sequence thereafter.

4.2.2.3 Military specifications and packaging data. - Military specifications and packaging data will be accepted on the basis of 100% examination by the Government for conformance with all requirements of this specification and supporting documents.

4.2.2.4 Classification of defects.

4.2.2.4.1 Critical defects. - A critical defect in technical data shall be defined as one, which if undetected prior to use of the technical data, might result in extensive damage to equipment and property, or injury to personnel during manufacture, installation, testing, transportation, or use of the item.

4.2.2.4.2 Major Defects - Category E - (Procurement) data. - A major defect in Category E technical data shall be defined as one which would contribute to one or more of the following deficiencies:

a. Inability to successfully procure or manufacture the item. Examples: Insufficient description; missing dimensions, tolerances, symbols, or other characteristics; non-existent part number; unavailable materials or processes; ambiguous or impossible manufacturing conditions or requirements.

b. Item manufactured or purchased in conformance with data will not assemble or function as intended. Examples: Incorrect mating dimensions; excessive tolerances; erroneous electrical characteristics; incorrect wire lists; improper selection of materials and finishes, no indication of subsequent fitting or selection, when intended at a later stage, etc.

c. Data will not support rejection of material, components or major items not suitable for their intended purpose. Examples: lack of specificity in detail requirements; lack of adequate inspection or test provisions; ambiguity or conflict between performance requirements and inspection provisions.

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d. Essential manufacturing and assembly sequence not specified. Examples: insufficient use of sub-assembly drawings, of section views, and assembly and adjustment instructions; omission of unique processing data essential to end product definition.

e. Technical data unduly limits competitive procurement. Examples: unauthorized use of proprietary components or processes; unnecessary use of "source control" drawings; specification control drawings unnecessarily "tailored" to a specific source; selection of materials and processes not readily available.

f. Technical data does not support maximum interchangeability of components. Examples: unnecessary use of selective assembly and "fit at assembly" techniques.

g. Errors in Engineering Parts Lists which can result in procurement or manufacture of incorrect quantities of components. Examples: Incorrect part numbers and quantity requirements; inconsistencies between parts list and assembly drawings or schematics.

h. Errors in Data List which can result in difficulty and error in compilation of a complete Technical Data Package. Examples: incorrect or omitted drawing numbers; incorrect or omitted specifications; failure to cross-reference supporting lists.

i. Inability to obtain a completely legible 4th generation reproducible contact print or a completely legible enlarged reproduction from second generation microfilm. Examples: insufficient density and contrast in original drafting or typing; insufficient spacing between lines; crowded and insufficiently large lettering; damaged or worn original documents.

4.2.2.4.3 Major Defects - Category G (Installation) data. - A major defect in Category G technical data shall be defined as one which would contribute to one or more of the following deficiencies:

a. Incorrect location of components

b. Incorrect wiring or piping installation

c. Incorrect mounting hardware

d. Incorrect or inadequate set-up and adjustment

4.2.2.4.4 Major Defects - Category H - (Maintenance) data. - A major defect in Category H technical data is one which would contribute to inability to successfully trace and diagnose the cause of malfunction, inability to properly disassemble and re-assemble the item, or inability to properly perform preventative maintenance. Examples: Insufficient

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or improperly selected assembled and sectional views; insufficient or erroneous schematics and diagrams; erroneous wire lists; incomplete or incorrect set-up and adjustment procedures; insufficient sub-assembly break-outs; errors in parts lists; no indication of lubrication requirements, etc.

4.2.2.4.5 Minor defects. - A minor defect in technical data shall be defined as any departure from the requirements of this specification and other supporting documents forming part of the contract which are not classifiable as critical or major defects. Included in this classification are:

a. Variations from standard drafting practice which do not materially affect the usefulness of technical data.

b. Variations from standard format requirements.

c. Errors in spelling; non-standard abbreviations; punctuation; and grammatical construction which do not materially affect the clarity of the document.

d. Departures from accepted standards of neatness and good workmanship which do not affect legibility and clarity of the document.

e. Use of non-standard components where suitable standards are available.

f. Use of unnecessarily restrictive tolerances, performance requirements, and materials, although manufacture is practical and competitive procurement is not unduly restrained thereby.

4.2.2.5 Disposition of rejected lots. - Rejected lots shall be screened by the contractor for all defective characteristics. After corrections have been made, the lot shall be resubmitted as a segregated lot, together with a report of corrective action taken. Failure of a lot to qualify for acceptance on a second examination will constitute prima facie evidence of the contractor's inability to perform under terms of the contract.

4.2.2.6 Disposition of individual non-conforming documents. - Rejected documents from accepted lots will be completely rechecked by the contractor and resubmitted as a single segregated document together with a report of corrective action taken.

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4.2.2.7 Schedule for government acceptance. - Processing time for acceptance or rejection of documentation, including official release after acceptance will be a maximum of 30 calendar days per 100 documents or fraction thereof.

4.3 Inspection provisions for prototype models.

4.3.1 Component inspection. - The contractor shall perform 100% inspection of all components on the basis of Government approved drawings and specifications. Unless otherwise authorized by the technical supervisor by means of an approved Request for Waiver (AMC Form 1020) or an approved Engineering Order, any variation between completed parts and specified requirements shall be cause for rejection.

4.3.2 Final inspection. - The contractor shall perform all inspection and tests required by the Government approved detail specification. Failure of the prototype to meet the functional and performance requirements of the specification, using the prescribed test methods and procedures, shall be cause for rejection of the prototype, pending correction of either the prototype or the technical data, as will be determined by the technical supervisor.

4.3.3 Government inspection. - The Government reserves the right to conduct inspection surveillance or independent inspection of any or all components and assemblies during the course of prototype manufacture and final inspection.

4.3.4 Contractor certification. - Prior to delivery of prototype models, the contractor shall certify that all components were manufactured or procured on the basis of Government approved drawings and specifications, and that no other technical data, Industry or company specifications which do not form a part of the final Technical Data Package were used to supplement the drawings and specifications during manufacture, purchase, or inspection. Any exceptions shall be cited in the certification (See 3.10.2) The contractor shall further certify that all components were given 100% inspection against the requirements of the Government approved drawings and specifications, and that any variations therefrom were authorized by means of an approved Request for Waiver or Engineering Order.

5. PREPARATION FOR DELIVERY.

5.1 Technical data. - Form 1 original drawings and Form 2 reproductions shall be carefully handled throughout the drafting, checking, and approval cycle, prior to delivery to the Government. Preparation for delivery shall be as prescribed in Specification MIL-D-1000.

5.2 Prototype model. - Packaging, packing and marking shall be in accordance with Level A packaging data developed by the contractor under provisions of 3.8 and approved by the Government, unless otherwise specified in the Technical Scope of Work or the contract.

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6. NOTES.

6.1 Intended use. - This specification is intended for use in conjunction with a detailed Technical Scope of Work which will delineate the specific objectives and performance requirements to be achieved on contracts for production engineering of Fire Control materiel. Where substantially less than half of this specification will apply to a contract, the applicable requirements of this specification should be extracted and made an integral part of the technical scope of work.

6.2 Ordering data.

6.2.1 Implementation of the requirements of this specification. - Use of this specification requires careful delineation in the Technical Scope of Work, or in the contract, of the applicability of each major requirement of this specification, any exceptions or additions thereto, and schedules for completion of each specific requirement. Recommended methods of referencing the requirements of the specification in the Technical Scope of Work are outlined in Appendix V.

6.2.2 Data requirements. - Department of Defense policy requires that data requirements of Defense contracts be specifically listed on a Contractor Data Requirements List, DD Form 1423, which will form a part of the contract. Appendix V provides a listing of all data requirements of this specification. Only those data which are essential to a justifiable intended use will be listed on the DD Form 1423.

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Project No. 1200-A018Y

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APPENDIX I

INSTRUCTIONS FOR THE PREPARATION OF SECTION 3 OF MILITARY SPECIFICATIONS FOR FIRE CONTROL MATERIEL

A. Scope

This Appendix to Specification MIL-F-46370 supplements Chapter V of Standardization Manual M200 for specific application to the preparation of Section 3 of Military Specifications for Fire Control Materiel.

B. General Requirements

Section 3 of a military specification shall be so written that compliance with all requirements will insure the suitability of the equipment for its intended purpose, and non-compliance with any requirement will indicate the unsuitability of the equipment for the intended purpose. Only those requirements shall be specified which are necessary and practically attainable.

C. Qualification

Qualification will not be specified as a requirement of Section 3 except as specifically directed by the technical supervisor. Establishment of a Qualified Products List (QPL) for a specific product restricts competition to those suppliers whose product has been tested and approved by the Government prior to procurement, and is justifiable only on the basis of time and expense involved in conducting special environmental, storage and reliability tests.

D. Preproduction or Initial Production Sample

The requirement for submission of a preproduction sample or initial production sample (first article submission) will normally be included in the production contract, rather than the detail specification. Nevertheless, in order to provide a basis for including appropriate quality assurance provisions in Section 4, the following requirement will be specified in Section 3:

"Requirements for the submission by the contractor of a preproduction (or initial production) sample shall be as specified in the contract."

A preproduction sample will normally be indicated in "detail performance" specifications in order to provide for design approval prior to undertaking full scale production. Detail performance specifications which include a qualification requirement (paragraph b, above) need not specify a preproduction sample, but may require an initial production sample inasmuch as the design has already been approved. An initial production sample will normally be indicated in a "detail design" specification, rather than a preproduction sample, inasmuch as the design has been specified by the Government.

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E. Independence of Section 3 requirements.

1. Each requirement of Section 3 shall be simply and concisely stated and umambiguous. Each requirement shall be self-sufficient and shall be stated in terms which are independent of test methods and procedures. Section 3 should provide a firm basis for preparation of Section 4. One test for adequacy of Section 3 is whether it would serve as a sound contractual basis for acceptance or rejection of equipment when detailed test methods and procedures are to be developed by the contractor.

2. The following is an example of a concise statement of the basic requirement which, while self-sufficient, permits and requires further expansion of the details of the test procedure.

> "Fungus Resistance - The equipment shall show no evidence of damage or deterioration of performance following an incubation period of 28 days after inoculation of all internal and external surfaces with a mixed spore suspension."

3. This statement of requirements leaves to Section 4 of the specification the identification of cultures to be used, the culture media, means of storage, method of preparing the suspension, the density count, method of applying the suspension to the equipment, and the incubation procedure.

4. A precise dividing line between Section 3 and Section 4 cannot always be drawn. In the above example, for instance, the identification of the cultures, the suspension density, and the temperature and humidity conditions during incubation might well have been part of the requirements statement. Their omission assumes that selection of the specific cultures and the density of the spore suspension can be left to the discretion of the contractor if omitted from Section 4, and that the maintenance of proper conditions of temperature and humidity are implicit in the word "incubation". If these assumptions can be made, the omission of such details in Section 3 is desirable from the viewpoint of simplicity and emphasis.

5. Specifically unacceptable as a statement of requirements are such statements as:

> "The equipment shall show no evidence of damage or deterioration of performance after being subjected to the fungus resistance test specified in paragraph 4.3.9."

Such a statement is not a self-sufficient requirement specification.

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F. Requirements for Preproduction, Initial Production, Qualification, and Special Samples

Specifications shall not differentiate between production items and special testing samples with respect to Section 3 performance requirements. All equipment to be delivered under the specification is expected to meet all of the requirements. Production items are expected to perform interchangeably with preproduction, initial production, qualification or special requirements. The fact that certain tests are conducted on a limited sampling basis, as prescribed in Section 4, has no effect on requirements specified in Section 3. Nevertheless, Section 3 requirements should be so specified as to be readily translatable to a practical test procedure. There shall be sufficient coordination between Section 3 and Section 4 specialists to insure that compliance with requirements will be determined by the most appropriate inspection procedure; i.e., 100% inspection, standard sampling per MIL-STD-105, or special sampling.

G. Avoidance of instructions and procedures

1. In order to clearly differentiate between test procedures and requirements, the specification of Section 3 requirements as instructions shall be avoided. Statements such as "Each item shall be subjected to....." or "the mean daily rate shall be computed....." are procedural instructions, not statements of requirements. The statement of requirements shall necessarily include the conditions relating to the performance requirement, but not to the methods and procedures by which these conditions are attained.

2. Section 3 requirements shall not be specified in terms of sampling or qualification procedures. The requirements as written shall be written to apply to all items supplied under the specification. It is the function of Section 4 to prescribe a reasonable sampling plan or qualification procedure.

3. The following examples illustrate the requirements of paragraphs 1 and 2 above:

a. Incorrect method:

"Long-Term Accuracy - For qualification, in addition to other requirements specified herein, ten watches shall operate at 75° ± 3°F for a total running time of between 180 and 200 days. Each watch shall be alternated between the dial-up and crown-down positions every seven days. At the conclusion of the total running time, the mean daily rate shall be determined for each watch. The average of the mean daily rates for nine watches shall not exceed 72 seconds, and no watch of the nine shall exceed a daily rate of 120 seconds."

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b. Correct method:

"Long-Term Accuracy - The average mean daily rate of watches supplied under this specification shall not exceed 72 seconds, nor shall the mean daily rate of any individual watch exceed 120 seconds over a period of at least 180 days of operation at a temperature of 75° ± 3°F, half of which time shall have been in a dial-up and half in a crown-down position, alternated at seven-day intervals."

c. These examples illustrate the importance of separating Section 3 requirements from Section 4 test methods and procedures. As stated in the first example, nine watches out of ten must meet certain accuracy requirements. Presumably every tenth watch need not be accurate. It also unintentionally establishes a requirement that the watches be capable of operating for a total running time of 120 days maximum, when it was actually intended that the 120 days be the limit of the accuracy test period only. For purposes of simplifying Section 3, the requirement that the position of the watches be alternated at seven-day intervals might preferably have been included in Section 4.

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APPENDIX II

INSTRUCTIONS FOR THE PREPARATION OF SECTION 4 OF MILITARY SPECIFICATIONS FOR FIRE CONTROL MATERIEL

A. Scope

This Appendix to MIL-F-46370 supplements Chapter V of Standardization Manual M200, for specific application to the preparation of Section 4 of Military Specifications for Fire Control Materiel. Its purpose is to facilitate the preparation of Section 4 through standardization of format, language, and selection of sampling methods. While some portions of this Appendix are specified to be mandatory, the recommendations and typical examples are not in any way intended to limit the exercise of best technical judgment in the preparation of Section 4, provided that variations do not conflict with Standardization Manual M200, or Department of the Army regulatory requirements.

B. Outline of Form

Section 4 of each specification shall be prepared in accordance with the following outline of form. Paragraph numbering and paragraph headings shall be as specified, except that numerical sequence of paragraphs shall be maintained in the event that neither preproduction nor qualification tests are applicable.

4. Quality assurance provisions
4.1 Responsibility for inspection
4.2 Preproduction or initial production test
4.2.1 Preproduction test
4.2.1.1 Preproduction sample failure
4.2.2 Initial production test
4.2.2.1 Initial production sample failure
(If Qualification is required, use the following)
(4.2 Qualification test)
(4.2.1 Qualification approval)
4.3 Inspection provisions
4.3.1 Submission of product
4.3.2 Examination and tests
4.3.3 Acceptance and rejection
4.4 Special sampling
4.4.1 Failure of sample
4.5 Inspection equipment
4.5.1 Government furnished inspection equipment
4.5.2 Contractor furnished inspection equipment
4.5.2.1 Government designs
4.5.2.2 Contractor designs
4.6 Test methods and procedures
4.6.1 etc. Sub-paragraphing as required

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C. Requirements

The requirements for preparation of Section 4 are specified below. For purposes of clarity and simplicity, requirements are prescribed under the appropriate paragraph number as listed above, in the outline of form. Where a specific instruction necessitates the use of more than one paragraph, alpha-numeric paragraphing is used to differentiate between the paragraphing of the instruction and the mandatory Dewey-decimal paragraphing of Section 4.

4.1 Responsibility for inspection. - Paragraph 4.1 of each specification shall delineate the general responsibilities of the supplier for performance of inspection for the Government, and shall reserve for the Government the right to perform such independent inspections as are deemed necessary. The language to be used in the introductory clause of paragraph 4.1 has been standardized for mandatory use in all specifications as follows:

> "4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements."

4.1.1 General provisions. - A subparagraph of 4.1 shall further delineate the requirements for inspection procedures and provisions for Government verification of components and subassemblies of fire control instruments standardized under MIL-F-13926 whenever mandatory detail requirements have been specified by means of Government drawings. Also to be included in this paragraph shall be the definition of inspection terms. The commonly used terms have been defined in MIL-STD-109, which should be cited in this paragraph in lieu of a compilation of definitions. A typical paragraph which will meet above criteria is as follows:

> "4.1.1 General provisions. - The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109."

4.2 Preproduction, initial production, or qualification testing.

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a. Preproduction or initial production test. - When a preproduction or initial production sample is specified in the contract, this paragraph shall specify whether that test shall be conducted by the Government, by the supplier, or both, the number of items to be tested and the characteristics to be inspected. Preproduction tests shall include all of the tests specified for acceptance inspection, in addition to those tests considered essential to demonstrate the adequacy of the contractor-developed design, selection of materials where optional, and such other features which, once established, need not be tested during normal production inspection. The initial production sample shall be representative of the materials, equipment, processes and procedures that will be used in regular production. A separate subparagraph, entitled "Preproduction sample failure", or "Initial production failure" shall specify the consequences of failure. The following paragraphs are considered typical examples of preproduction or initial production test requirements, and action upon failure of sample:

"4.2 Preproduction or initial production test.

"4.2.1 Preproduction test. - When specified in the contract, one preproduction sample shall be submitted to the procuring agency for approval of design and construction features. Preproduction samples shall have successfully met all Inspection Provisions of Paragraphs 4.3 and 4.6 and shall then be inspected in accordance with the following additional tests:

SPECIAL CHARACTERISTICS	REQUIREMENTS	TEST METHODS AND PROCEDURES

(Tabulation of each title of test (characteristic), the applicable "requirement" paragraph number of Section 3, and the applicable "test methods and procedures" paragraph number of Section 4 for all special preproduction requirements and tests.)

"4.2.1.1 Preproduction sample failure. - Should one preproduction sample fail to meet the constructional and performance requirements of this specification, a corrected sample shall be resubmitted. Failure of the resubmitted sample to meet the requirements of this specification will constitute prima facie evidence of the contractor's inability to perform under terms of the contract.

"4.2.2 Initial production test. - Initial production test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, samples shall be the first two production units and shall have been manufactured using the same materials, processes and procedures as will be used in regular production.

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Initial production testing shall be in accordance with 4.3 and 4.6 and shall be witnessed by the Government representative for all examinations and tests. The Government reserves the right to conduct any or all examinations or tests as may be deemed necessary.

"4.2.2.1 Initial production sample failure. - Should one unit fail to meet the initial production testing specified in 4.2.2, the contractor shall cease those operations which are causing the rejection and make the necessary corrections thereto. When corrections in the process or controls have been made, two additional (new) production models shall be submitted in accordance with 4.2.2."

b. Qualification tests.

When a requirement for Qualification Testing is specified in Section 3 of the specification, this paragraph shall specify the number of items to be submitted for qualification, and the characteristics to be inspected. It shall require the submission of data to show that the manufacturer maintains an adequate and effective quality assurance system, that he owns or has the use of necessary test equipment, and that he is capable of delivering acceptable items within a reasonable procurement lead time. It shall further specify that testing will be performed in a Government Laboratory by Government personnel upon receipt of qualification specimens accompanied by a report of successful test and certification of compliance with the Qualification Requirements of Section 3. Qualification testing shall include all of the tests specified for acceptance inspection, in addition to those tests considered essential to demonstrate the adequacy of the contractor-developed design, selection of materials where optional, and manufacturing techniques. A separate subparagraph entitled "Qualification Approval" shall specify the conditions under which approval will be granted or refused; a typical example of a Qualification Test requirement is as follows:

"4.2 Qualification test. - Six specimens conforming with the requirements of Section 3 of this specification shall be submitted to the designated Government Laboratory (See 6.2) for testing by Government personnel. The specimens shall be accompanied by:

a. A certification that the construction, materials, and workmanship used in fabrication of the specimens are representative of the items which would be offered under this specification.

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b. A list of all test equipment utilized to perform required tests.

c. A written description of the quality assurance system used in the manufacturer's normal operations.

d. A summary of operations and tests performed on sub-contract, and identification of the sub-contractor.

e. A report attesting to the successful compliance with all of the Inspection Provisions prescribed in Paragraphs 4.3 and 4.6 of this specification and the following additional qualification tests:

SPECIAL CHARACTERISTICS	REQUIREMENTS	TEST METHODS AND PROCEDURES

(Tabulation of title of test (characteristic), the applicable "requirement" paragraph number of Section 3, and the applicable "Test methods and procedures" paragraph number of Section 4, for all special Qualification requirements.)

"4.2.1 Qualification approval. - Potential suppliers who have submitted specimens for Qualification Testing will be placed on the Qualified Products List (QPL) for this specification at such time as the specimens have successfully passed all of the required tests in a Government Laboratory. Failure to meet all of these requirements will result in failure to obtain Qualification Approval. Qualification specimens will be accepted for retest upon presentation of satisfactory evidence that deficiencies have been corrected. The Government reserves the right to conduct such on-site inspections on the potential supplier's premises as may be required to insure that the manufacturing processes and quality control measures used in fabrication of the specimens are indicative of the quality of the item which may be expected in production."

4.3 Inspection provisions. - This paragraph shall summarize the requirements for submission of product, examination and tests, classification of defects, Acceptable Quality Level, inspection level, acceptance and rejection pertaining to inspection of regular production under Government supply contracts. The content of this paragraph shall be divided into the following subparagraphs:

4.3.1 Submission of product.

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4.3.2 Examination and tests (Including reference to tables of tabulated classification of defects, AQLs and sampling plans).

4.3.3 Acceptance and rejection

4.3.1 Submission of product. - This paragraph shall incorporate the "Submission of Product" provisions of MIL-STD-105 by reference. The following standardized language will be generally suitable:

"4.3.1 Submission of product. - Unless otherwise specified by the Contracting Officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105."

4.3.2 Examination and tests. - This paragraph shall define the amount of inspection to be conducted in determining compliance with the requirements of Section 3 and Section 5 of the specification as well as of drawings and other technical data forming a part of the specification. Examination provisions for requirements specified on drawings are contained in MIL-F-13926, the requirements of which shall be included in this paragraph by reference. Examination provisions for the requirements of Section 3 and 5 shall be specified by means of utilizing MIL-STD-105 and delineating Classification of Defects tables showing Acceptable Quality Level (AQL) and test procedure for each characteristic to be examined. When "Special" Inspection Levels in MIL-STD-105 is to be applied, the special level shall be stated.

a. Criteria for specifying Classification of Defects (CDs), Acceptable Quality Level (AQL), and sample size:

(1) Classification of defects

Each requirement (characteristic) specified in Section 3 and 5 of the specification which is intended to be subject to examination prior to acceptance shall be classified as to the effect of any failure to comply with the specified requirement. Such defects shall be classified as Critical, Major, or Minor, as defined in MIL-STD-105. All characteristics shall be tabulated by their title, the applicable "requirement" paragraph number, and the "test methods and procedures" paragraph number, grouped in accordance with their classification. Characteristics shall be numbered in sequence within their classification group. Critical defects shall be numbered from 1 through 99; Major defects from 101 through 199; Minor defects from 201 through 299. If more than one Classification of Defects table is necessary because of different inspection levels, sampling plans, or AQLs, serial numbers within each classification group shall be sequentially numbered throughout the specification, irrespective

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of the number of tables required; e.g.: if the last major characteristic in Table I is numbered 107, then the first major characteristic in Table II shall be numbered 108.

(2) Acceptable Quality Level (AQL)

Each Classification of Defects shall be assigned an Acceptable Quality Level on a single defect (individual characteristic) basis, in accordance with the definition in MIL-STD-105. Major defects shall normally be assigned an AQL of .65%, although an AQL of up to 1.5% may be specified where appropriate. Minor defects shall normally be assigned an AQL of 2.5%, with a maximum limit of 6.5%, if appropriate. If for any substantial reason, it should be necessary to assign two or more different AQLs to the same classification of defects, separate tabulation of CDs will be required.

(3) Inspection Level

The sample size and appropriate inspection level for each classification of defects, except critical defects, shall be in accordance with "Sampling Plan" and "Inspection Level" as specified in MIL-STD-105. A smaller sample size, as provided by "Special" inspection level S-1 through S-4 in MIL-STD-105 may be specified for certain expensive, time-consuming, or destructive tests such as environmental tests. When other than a normal inspection level (Level II) is desired, the inspection level shall be specified immediately below the heading; e.g.:

TABLE I - CLASSIFICATION OF DEFECTS
USE SPECIAL INSPECTION LEVEL S-4 IN TABLE I, MIL-STD-105

Special sampling plans not available in MIL-STD-105 may be used where relatively small sample sizes are necessary and large sampling risks can or must be tolerated. Critical defects shall be subject to 100% inspection unless otherwise specified. Certain major defects may be selected for 100% inspection if practical; usually in cases where automated inspection equipment will be used, numerous individual adjustments or calibrations are necessary in the major item; or, on expensive major items of equipment, where it is evident that 100% inspection would cost only a small fraction of production costs. All characteristics intended to be inspected 100% shall be separately tabulated. A separate tabulation of CDs shall be shown for special sampling plans not available in MIL-STD-105.

b. Examples illustrating the requirements of an acceptable "Examination and test" paragraph:

"4.3.2 Examination and tests. - Components and sub-assemblies shall be inspected in accordance with the inspection provisions contained in MIL-F-13926, and the Supplementary Quality Assurance Provisions (SQAP) listed in the List of Inspection Documents, IEL_____. Examination and tests related to Section 3 herein

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shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105, and the sampling plans specified in Table(s)_____ herein. Examinations and tests for Packaging, Packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the Supplier prior to Government acceptance or rejection by lot."

c. Example showing Format of Classification of Defects Table:

TABLE I - CLASSIFICATION OF DEFECTS

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE		
MAJOR: AQL .65% Defective		
101. Balance	3.1.4	4.5.6
102. Repeatability	3.1.5	4.5.2
103. Etc.		
MINOR: AQL 2.5% Defective		
201. Level	3.1.1	4.5.3
202. Hinge Stops	3.2.3	4.5.10
203. Etc.		

4.3.3 Acceptance and rejection. - Each specification shall include provisions for the screening of rejected lots to eliminate defectives, and for the subsequent resubmission of such lots. The following standardized language is considered generally suitable:

"4.3.3 Acceptance and rejection. - Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105."

4.4 Special sampling. - Under conditions requiring a preproduction or qualification test, it is often desirable to repeat all or a portion of those tests during production, in addition to normal inspection required by paragraph 4.3, in order to insure that no significant change

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has occurred in construction, materials, or critical manufacturing processes since approval of the preproduction or qualification sample. Since these tests are usually costly, time-consuming, or require special laboratory or field equipment, they are normally conducted on a small sampling basis. The sampling may be specified as a low percentage of production, or a small sample from each month's production whichever occurs first. Special sampling may also be specified for certain time-consuming or costly regular inspection tests in lieu of sampling in accordance with MIL-STD-105, if appropriate (See 4.3.2a(3)). When it is determined that special sampling is required, the essence of the following paragraph shall be specified.

"4.4 Special sampling. - One (item) shall be selected at random from each 100 produced, or from each month's production, whichever occurs first. The sample shall have successfully met all other examinations and acceptance tests specified herein, and shall meet the following requirements and tests:

CHARACTERISTIC REQUIREMENT TEST PROCEDURE

4.4.1 When more than one special sample size is desired for different types of testing, the essence of the following shall be specified:

"4.4 Special sampling.

"4.4.1 General. - One (item) shall be selected at random as a special sampling from each 100 produced, or from each month's production, whichever.....

"4.4.2 Temperature. - Three (item) shall be selected at random as samples from each 50 produced or from each month's production, whichever.....

"4.4.3 Shock and vibration. - Three (item) shall be selected at random as samples from each 50 produced or from each .....etc.

"4.4.4 Failure of sample. - Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor, and resubmitted samples have been approved (See 4.3.3).

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4.5 Inspection equipment. - This paragraph shall define the supplier's responsibility for supply and maintenance of inspection equipment required to determine conformance with the requirements of the specification. These responsibilities have been delineated in MIL-I-45607 and shall be incorporated into this paragraph by reference to the specification. The following has been standardized for recommended use:

"4.5 Inspection equipment. - Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607."

4.5.1 Government-furnished inspection equipment.- This subparagraph shall specify the contractor's responsibility for the care and maintenance of any Government Furnished Test Equipment provided under terms of the contract. These responsibilities are defined in MIL-I-45607 and shall be incorporated by reference, as follows:

"4.5.1 Government furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607."

4.5.2 Contractor-furnished inspection equipment. - This paragraph shall include requirements governing the design and construction of inspection equipment by the contractor. Where mandatory Government designs are to be included in the specification, the contractor shall be required to supply test equipment conforming with these requirements, or an approved equal alternate in accordance with the applicable requirements of MIL-I-45607. The following has been standardized for recommended use in identifying inspection equipment lists (List of Inspection Documents) and specifying contractor design responsibilities:

"4.5.2 Contractor furnished inspection equipment.

"4.5.2.1 Government design. - Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract, shall be supplied by the contractor in accordance with technical data listed in List of Inspection Documents, IEL_____.

"4.5.2.2 Contractor design. - The contractor shall design and supply inspection equipment compatible with "Test Methods and Procedures" specified in Paragraph 4.6 of this specification, and with the "Component Inspection" requirements of MIL-F-13926.

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Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of the test equipment."

4.6 Test methods and procedures.

a. General

The preceding paragraphs of Section 4 have been amenable to standardization of content, and to a great extent, the specific language to be used. Such is not the case with the development of "Test Methods and Procedures". Like the initial development of Section 3 requirements, the specification of meaningful and useful Test Methods and Procedures demands utilization of the best available engineering and related technical skills. Each requirement specified in Sections 3 and 5 of the specification shall be carefully analyzed, and a determination made as to the methods, procedures, equipment, and sequence of inspections and tests that will best insure the acceptance of those products which meet the requirement, and the unequivocal rejection of those which do not conform.

b. Extent of Coverage

Each separate requirement of Sections 3 and 5 shall be covered by a description of the test method and procedure to be used to inspect for conformance. Certain extremely simple and obvious procedures which can be described by one word or phrase such as "gage", "scale", "visual", etc. may be shown in the Classification of Defects tables in lieu of a "Test Methods and Procedures" paragraph, provided that no special instructions, precautions or equipment descriptions are needed. In general, the test methods and procedures will be specified in the same order as the requirements are specified in Section 3. It is more important, however, that they be grouped in a logical test sequence, particularly when a number of requirements are to be tested under the same test set up.

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c. Use of general Specifications

Where standardized test methods and procedures are established in general military specifications and military standards, they may be utilized by reference thereto in Section 4 and listed under Section 2 of the detail specification. Some general specifications most commonly used for fire control materiel are MIL-F-13926, MIL-O-13830, MIL-STD-105, MIL-STD-202, etc. Where similar test methods and procedures exist in more than one specification or standard, the precedence to the document shall be based upon consistency in use in previously prepared detail specifications for fire control materiel. For example, if environmental tests are similar in MIL-F-13926, MIL-STD-202 and MIL-E-5272 then the use of MIL-F-13926 for fire control materiel is preferred. In general, however, the most technically suitable existing document will be used.

d. Test Equipment

(1) Unless the test method is to be based on a specific mandatory test equipment design, it is neither necessary nor desirable to envision a single integrated test set up; the equipment for each test shall be described independently of any other test. Where an integrated testing device has been developed and will be mandatorily specified, the test methods and procedures shall preferably be grouped in the sequence of operations required for effective utilization of the special testing device.

(2) The determination of the number and types of mandatory test equipment designs to be incorporated in "Test Methods and Procedures" if any, shall be made jointly by the specification engineer and the test equipment designer based on criteria specified in "Inspection Equipment" requirements of the specification of which this Appendix is a part. In accordance with these criteria, test methods and procedures shall normally be initially developed on the assumption that no test equipment designs other than Class A and B test equipment will be specified as a Government design. Class C or E test equipment shall be specified as mandatory only when such designs have been completed, tested, and are available for release concurrently with release of the specification.

(3) General instructions for the set up of mandatory test equipment which are included on the test equipment drawing, shall be made a part of the appropriate "Test Methods and Procedures" paragraph of the specification by reference to the drawing. Operation of the test equipment shall be appropriately defined in the respective test methods and procedures written in the specification for the requirements to be tested.

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APPENDIX II (Cont'd)

(4) Test equipment not specified as a mandatory Government design, shall be described verbally or schematically in such detail as is necessary for intent and clarity of the test method.

(5) Commercial model numbers shall not be used to augment a test equipment description. If the commercial model number is an essential part of the description, the item shall be considered as Class A test equipment, and shall be covered by a Government "Source Control" or "Specification Control" drawing.

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APPENDIX III

INSTRUCTIONS FOR THE PREPARATION OF SUPPLEMENTARY QUALITY ASSURANCE PROVISIONS (SQAP)

A. Scope

This Appendix to Specification MIL-F-46370 provides instructions for the preparation of Supplementary Quality Assurance Provisions (SQAP) sheets as required by Paragraph 3.3 of the basic specification.

B. General Requirements

General quality assurance provisions for Fire Control components are contained in MIL-F-13926 and MIL-F-45133. Certain components of Fire Control materiel, selected in accordance with criteria contained in the basic specification, require the preparation of SQAP sheets which specify detailed quality assurance provisions in a manner essentially similar to Section 4 of a military specification. A SQAP bears the same relationship to the engineering drawing that Section 4 of a military specification bears to the requirements specified in Section 3 of the specification.

C. Outline of Form

1. Each SQAP shall include the following introductory note: "THIS SQAP FORMS A PART OF SPECIFICATION MIL-F-13926 OR MIL-F-45133 AS SPECIFIED IN THE CONTRACT. INSPECTION SHALL BE CONDUCTED AS SPECIFIED HEREIN, AND IN ACCORDANCE WITH REFERENCED DOCUMENTS."

2. The SQAP shall consist of four parts as follows:

PART	I	Inspection Provisions
PART	II	Certification Provisions
PART	III	Test Methods and Procedures
PART	IV	List of Inspection Documents

3. Part I - Inspection Provisions

a. Minimum criteria for inspection of components and sub-assemblies shall be presented in the form of one or more tabulations identified by Roman numerals (Table I, II, III, etc.) and one of the following titles, as appropriate:

(1) Classification of Defects

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APPENDIX III (Cont'd)

(2) 100% Inspection

(3) Special Sampling

b. Classification of Defects and 100% Inspection Tables

(1) Examination criteria shall be tabulated under appropriate column headings and shall include each characteristic number as shown in the following example;

CLASS	CHARACTERISTIC	PER DWG OR SPEC	INSPECTION METHOD
CRITICAL:	NONE		
MAJOR:			
101.			
102.			

(2) Sampling Instructions

Each table titled "Classification of Defects" shall contain the following instruction below the title: "Use Sampling Plan Tables in MIL-F-13926 or MIL-F-45133 as applicable."

(3) Class

Each requirement or characteristic specified on the applicable drawing which is intended to be subject to inspection prior to acceptance shall be classified as to the effect of any failure to comply with the specified requirement. Such defects shall be classified as Critical, Major, or Minor, as defined in MIL-STD-105. Critical Defects, if any, shall be tabulated first, followed by tabulation of Major Defects, than Minor Defects, in that order. Tabulations of Critical or Major Defects which require "Certification" of materials or processes, in lieu of verification, shall be included in Part II of the SQAP, as applicable.

(4) Numbering of Defects (characteristics)

Defects shall be numbered sequentially within their classification group. Critical defects shall be numbered from 1 through 99; Major defects from 101 through 199; Minor defects from 201 through 299. The appropriate number of each defect (characteristic) shall appear in column form below the "Class" heading (see example in (1) above). If both

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a "Classification of Defects" table and a "100% Inspection" table are required, the defect (characteristic) numbers representing each classification group shall be sequentially numbered from table to table; e.g. if the last Major characteristic of Table I is numbered 107 the first Major characteristic of Table II shall be numbered 108 (unless classified as "Critical", characteristic numbering for 100% inspection shall be as prescribed for Major defects). Sequential numbering will prevent a characteristic number (101, 201, etc) from being repeated in any one SQAP.

(5) "Characteristic" Column

Description of characteristics shall be as brief as possible, but sufficiently definitive to clearly identify the characteristic from other characteristics. The less significant characteristics may be grouped under a single characteristic description such as "completeness of machining operations", or "completeness of assembly", and classified as "minor".

(6) Drawing or Specification Number Column

The "Per Dwg or Spec" column of the Inspection Provisions table is intended for use on SQAP's applicable to sub-assemblies where certain characteristics of components of the sub-assembly are subject to inspection. In such cases, the number of the drawing or specification upon which the selected characteristic is specified will be specified in this column.

(7) "Inspection Method" Column

This column is intended to define either directly, or by reference, the examination, inspection methods and procedures by which each requirement will be inspected. Simple and obvious procedures may be included in the table by single words or brief phrases such as "visual", "tactile", "gage", "scale", etc; or by reference to an existing drawing or other document which describes the inspection methods. These brief phrases or references shall not be used, however, if any special instructions, precautions, or test equipment descriptions are needed. In such cases, reference will be made to the appropriate paragraph of Part III of the SQAP which will specify the test methods and procedures to be used.

c. "Special Sampling" Table

Special sampling plans may be specified in lieu of a standard lot sampling plan for certain unusually costly, destructive, or time consuming tests. The sampling may be specified as a low percentage of production, or a small sample from each month's production, whichever occurs first. When it is determined that special sampling is required,

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the tabulation of Inspection Provisions shall include in the heading the essence of the following sampling instructions, using an appropriate sample size:

> "Special sampling. - Three (item) selected at random from each 50 produced, or one from each month's production, whichever occurs first, shall be subjected to the following:

NO.	CHARACTERISTIC	PER DWG OR SPEC	INSPECTION METHOD

Each characteristic shall be numbered in sequence starting with the number 301. When tests are specified and tabulated as above, the following statement shall be used: "Failure of any one sample shall be cause to cease those operations causing failure until corrections have been made and approved by the Government. Sampling shall not be resumed until_____consecutively produced_____item(s) are accepted by the Government.

4. Part II - Certification Provisions

a. Where materials, heat treatment, finishes, welding, brazing, bonding, or other processes are of critical or major importance to the function of the item, but are not amenable to examination under normal inspection of the completed item, a requirement for certification shall be specified in Part II of the SQAP.

b. The following certification requirement shall be stated immediately below the Part II heading:

> "Certification shall include test data and results for each characteristic specified below. Certification shall be provided prior to performing final acceptance inspection, and will suffice for the remainder of the contract provided the manufacturing processes and techniques used to produce the items for which certification was issued have not been changed. Any and all changes will require a new certification from the contractor."

c. Certification requirements shall be tabulated under the following headings:

NO.	CHARACTERISTIC	TEST DATA TO COMPLY WITH	CERTIFICATION METHOD

(1) Each characteristic shall be numbered in sequence, starting with number 401, unless there is no "Special Sampling" table preceding, in which case the first characteristic shall be numbered 301.

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APPENDIX III (Cont'd)

(2) The characteristic to be certified shall be briefly identified as "weld", "bond", "heat treat", etc.

(3) Under "Test Data to Comply With", specify the document which contains the test requirements.

(4) The Certification Method columns will reference the paragraph of Part III which specifies the basis and form of the certification to be provided. Where no reference to Part III is required, this column may be deleted.

d. Where no certification is required, the words "none required" shall be added under the heading Part II - Certification Provisions.

5. Part III - Test Methods and Procedures

a. General

Each characteristic listed in Part I and Part II of the SQAP shall be carefully analyzed, and a determination made as to the methods, procedures, equipment, and sequence of examinations and test that will best insure the acceptance of those products which conform with requirements, and the rejection of those which do not conform. To insure correlation of the references under the "Inspection Method" column in Part I, the following parenthetical remarks shall be entered below the Part III title: "(As referenced under Inspection Method column, Part I)."

b. Extent of Coverage

Except for those simple and obvious procedures which can be described by a single word or a brief phrase, or by reference to an existing drawing or other document in the tabulation of Inspection Methods, each characteristic listed in Part I and Part II of the SQAP, shall be covered in Part III by a detailed description of the Test Method and Procedure to be used to inspect for conformance.

c. Test Equipment

The determination of the number and types of mandatory test equipment designs to be incorporated in "Inspection Methods" if any, shall be made jointly by the SQAP writer and the test equipment designer based on criteria specified in "Inspection Equipment" requirements of the specification of which this Appendix is a part. In accordance with these criteria, test methods and procedures shall be initially developed on the assumption that no test equipment designs other than Class A and B test equipment will be specified as a Government design. Class C or E test

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equipment shall be specified as mandatory only when such designs have been completed, tested, and are available for release concurrently with the SQAP. All other required test equipment shall be described verbally or schematically in such detail as is necessary for clarity of the test method. Commercial model numbers shall not be used to augment a test equipment description. If the commercial model number is an essential part of the description, the item shall be considered as Class A test equipment and shall be covered by a Government "Source Control" or "Specification Control" drawing.

d. Paragraph Numbering

Inspection Methods referenced in Parts I and II will be defined in Part III and will be numbered in numerical sequence starting with paragraph 1. Sub-paragraphs applicable to a single characteristic will be numbered by the Dewey-decimal system, e.g.: 1.1, 1.2, etc.

6. Part IV - List of Inspection Documents

a. This part of the SQAP shall comprise a listing of technical data applicable to inspection requirements for the pertinent component or sub-assembly. Format for Part IV shall be as follows:

PART IV - LIST OF INSPECTION DOCUMENTS

LIST OF DRAWINGS

NOMENCLATURE	DRAWING NO.	DATA LIST

LIST OF SPECIFICATIONS

TITLE	NUMBER

LIST OF STANDARDS

TITLE	NUMBER

b. The "Data List" column shall be used to indicate that the specified drawing contains a subordinate list of drawings. When the "Data List" column is used, the following note should be added after the last drawing listed:

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APPENDIX III (Cont'd)

"An entry in the Data List column indicates that additional drawings listed on the specified drawing form a part of this List of Inspection Documents."

c. When a drawing comprises more than one sheet, the number of sheets shall be shown parenthetically after the drawing number.

D. Other Format Requirements

1. SQAP shall be prepared on SMUFA Form 3186 which will be furnished by the Government upon request.

2. SQAP Number

3. Title Blocks - The Title Block of a SQAP shall include the words "Supplementary Quality Assurance Provisions For" followed by the nomenclature of the item to which the SQAP applies.

4. Code Identification Number - The lower right hand portion of the open area in the body of the SQAP shall contain the following: "CODE IDENT. NO. 19200".

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APPENDIX IV

INSTRUCTIONS FOR THE PREPARATION OF PACKAGING DATA SHEETS

A. Scope

This Appendix to Specification MIL-F-46370 provides instructions for the preparation of Packaging Data Sheets as required by paragraph 3.8 of the basic specification.

B. General Requirements

1. Preliminary draft. - A preliminary draft of each Packaging Data Sheet prepared on AMC Forms 1029 and 1029-1 as required, shall be submitted to the technical supervisor for review and approval.

NOTE: AMC Form 1029-1 shall be used for continuation of notes, sketches or illustrations

2. Final copy. - The final copy of the packaging data sheet shall be prepared on the basis of the approved draft and comments received from the technical supervisor. Preparation shall be accomplished on multilith masters of packaging data sheets using Elite Type and multilith ribbons. Illustrations shall be rendered in multilith pencil or transferred to a multilith mat by suitable printing methods. Care shall be exercised to prevent typewritten information from touching preprinted lines, numbers, or titles.

3. Notes. - When requirements are entered in "Notes", they shall be identified by asterisks; e.g., *, **, ***. Additional notes contained on continuation sheets shall be identified by letters A, B, C etc.

4. Test. - Coverage of the packaging data sheets shall include disassembly, cleaning, preservation, packaging, packing, and marking; the use of adequate protective measures to prevent deterioration resulting from exposure to atmospheric conditions during storage and shipment; appropriate wrappings, barrier and cushioning material; adequate bracings, blocking, cushioning, weather proofing and strapping.

NOTE: Disassembly of an item is often desirable to reduce the cube of the package for an item; however, the contractor shall obtain the approval of the technical supervisor prior to specifying disassembly on the Packaging Data Sheet.

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APPENDIX IV (Cont'd)

5. Dimensions. - Dimensions involving mixed numbers shall be recorded on the Packaging Data Sheet as a whole number first, a space, and then the fraction; e.g., 11 1/4. All fractions shall be specified as a number (the numerator) a virgule (/) and another number (the denominator), e.g., 1/8, 5/8, etc. Fractions provided by a single key of the typewriter may be used.

6. Weights and cubes. - Weights and cubes shall be specified decimally to the nearest hundreth (.01) pound or thousandth (.001) cubic foot for the item, unit package, and intermediate package logistics. Items of less than .01 pound or .001 cubic foot shall be specified as .01 or .001. Weights for exterior containers shall be specified to the nearest pound; volume shall be specified to the nearest hundreth (.01) of a cubic foot. A decimal point shall follow whole numbers.

7. Illustrations. - Where illustrations are necessary to supplement the text of the instruction, they shall be rendered isometrically and shall include call-outs and materials required. Illustrations will be prepared on continuation sheets, AMC Form 1029-1.

8. Engineering drawings. - Detailed engineering drawings, when required, shall be referenced on the Packaging Data Sheet.

C. Detailed requirements.

1. Federal item name. - Enter the Federal item name (FIN) for the assigned materiel. The entry should provide adequate item identification for marking purposes. When absolutely necessary, the nomenclature may be continued in "Notes"; however, the description is preferably restricted to the space provided.

2. Part number. - Enter the complete part number of the item.

3. Federal stock number. - Enter the complete Federal stock number.

4. Composition. - Enter the composition of the item being packaged; e.g. steel, brass, steel and rubber, plastic, wood, etc.

5. Finish. - Enter the type of protective finish of the item being packaged; e.g. paint, phosphate, chromeplate, black oxide, anodize, zinc plate, etc. When no finish is specified in the item requirements, the word "None" shall be entered in this block.

6. SNL or class. - Leave blank.

7. Technical service stock number. - Leave blank.

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APPENDIX IV (Cont'd)

8. Logistics data (Level A - Level B). - The following instructions are applicable to both Level A and Level B with respect to completion of the packaging data sheets, and both levels will be completed throughout. When Levels A and B are the same, Level B requirements shall be left blank and the words "Same as Level A" inserted in a staggered manner or stamped diagonally, in the center of all size columns, as applicable, under Level B.

a. Item. - Enter the unpreserved, bare dimensions in inches and fractions thereof of the item to be unit packaged. If disassembly is required prior to packaging, the dimensions taken shall be those of the item in a disassembled state arranged in the order that it will be packaged. In the event the item to be packaged requires that it be rolled, coiled, or folded prior to packaging, the dimensions will be those of the item in its rolled, coiled, or folded condition. Citation will be made to "Notes". Enter the actual weight of the item to be included in the unit package. Decimals recorded shall be in accordance with the requirements of B(6) above. Cylindrical volume shall be specified if applicable.

b. Unit package. - Enter the quantity of items, exterior size, gross weight, and cube of one complete unit package. The cube for cylindrical packages shall be computed on the basis of rectangular solids. Decimals shall be specified in accordance with the requirements of B(6) above. When single items are packaged and packed as a single unit, i.e.: the unit package and the exterior container are one and the same, the weight, dimensions, cube, etc., specified on the data sheet will be the same.

c. Intermediate package. - Enter the quantity of unit packages, exterior size, gross weight, and cube of one complete intermediate package. The volume for cylindrical packages shall be computed on the basis of rectangular solids. Decimals shall be specified in accordance with the requirements of B(6) above.

d. Exterior pack. - Enter the quantity of items, exterior size, gross weight, and cube of one complete exterior pack. The cube for cylindrical packs shall be computed on the basis of rectangular solids. Decimals shall be specified in accordance with the requirements of B(6) above.

9. Packaging requirements. - The following instructions are applicable for packing and packaging to both Level A and Level B with respect to completion of the Packaging Data Sheet.

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APPENDIX IV (Cont'd)

a. Method. - Enter the unit protection (submethod) after "Method", which is applicable to the particular item being packaged. Methods and sub-methods used shall be in accordance with MIL-P-116.

b. Cleaning. - Enter the applicable cleaning process(es) specified in MIL-P-116, by symbol.

c. Drying. - Enter the applicable drying procedure(s) of MIL-P-116 by symbol.

d. Sequence of operations (packaging requirements). - The unit packaging operations shall be numbered after the name in the sequence in which the work is performed.

e. Preservative. - Enter the preservative specification(s) or preservative type(s) as specified in MIL-P-116.

f. Intimate wrap. - Enter the applicable material by specification number, size, style, type, grade, and/or class.

g. Cushioning. - Enter the applicable material by specification number, size, style, type, and/or class. If clarification is required regarding the use of the material, specific instructions shall be provided in "Notes". In cases where special die cuts, filler pads, inserts, etc., are required, AMC Form 1029-1 shall be used for implementing requirements (sketches, photographs, bills of material, etc.). Any necessary supplementary drawings shall be referenced on AMC Form 1029-1.

h. Unit container. - Enter the specification, size, style, type, grade, and class, using symbols and numbers as reflected in the applicable specification.

i. Closure. - Enter the specification, size, type, grade, and class of tape; or type, grade, and class of adhesive, to be used in completion of the unit package.

j. Blank spaces. - The three blank spaces are to be used for listing additional materials that are normally used in packaging; e.g., desiccants, humidity indicator, blocking, etc., which will be used in completion of the unit package.

k. Intermediate container. - Enter the specification, size, style, type, grade, and class, using symbols and numbers as reflected in the applicable specification.

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APPENDIX IV (Cont'd)

1. Exterior pack; type of load. - Enter the "Type of Load" (1-Easy, 2-Average, 3-Difficult) as classified in JAN-P-100.

m. Exterior container. - Enter the specification number, size, style, type, grade, and class of the preferred container. Optional containers may be specified in the remaining three blank spaces. Cost, weight, and cube factors should influence the order in which the optional containers are listed.

10. Level C package and pack. - When Level C packaging and packing are required, specific Level C requirements will be entered on the packaging data sheet.

11. Notes. - Enter any implementing requirements, explanatory remarks, and necessary information for which insufficient space or no space at all is provided elsewhere on the Packaging Data Sheet. When additional space is necessary AMC Form 1029-1, continuation sheet, will be used.

12. Preparing activity. - Enter name or symbol of the mission activity for the Packaging Data Sheet. (FA)

13. Date. - Enter the date on which the Packaging Data Sheet was typed.

14. Individual packaging data sheet number. - The number for the packaging data sheet is derived by combining the part number prefixed by a capital gothic letter "P" with the controlling military specification number; e.g. MIL-P-14232/P6224055.

15. Sheet number. - Enter in proper sequence the number of the packaging data sheet.

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APPENDIX V

IMPLEMENTATION OF REQUIREMENTS OF SPECIFICATION MIL-F-46370A(MU)

A. Scope

This Appendix provides guidance for implementing the requirements of Specification MIL-F-46370 in the Technical Scope of Work of the contract and provides a listing of all technical data requirements of the specification for use in compiling Contractor Data Requirements List, DD Form 1423.

B. Referencing Specification MIL-F-46370 in the Technical Scope of Work

1. The requirements of this specification may be included in the detail Technical Scope of Work under the heading of "General requirements". It is recommended that General Requirements constitute the first requirements paragraph under Section 3 of the Technical Scope of Work, as in the following example:

> "3.1 General requirements. - The following requirements of Section 3, Specification MIL-F-46370, in their entirety, form a part of the requirements of this Technical Scope of Work, unless otherwise noted:
>
> (List each applicable contractor performance requirement by major paragraph title and number, as listed in 1.2 of this specification, indicating any exception within the major requirement paragraph by sub-paragraph title and number, or by other brief statement of exception. Example:
>
> a. Product Engineering Studies (3.1)
>
> b. Engineering Drawings (3.2)
>
> c. Inspection Equipment (3.5) Class A only.
>
> d. Engineering Lists (3.6)
>
> e. Data Approval and Release (3.9, except 3.9.3)"

2. It is recommended that the first sub-paragraph of General Requirements be used to specifically list those major requirements of the specification which are not required by the Technical Scope of Work. Example:

> "3.1.1 Non-applicable requirements. - The following requirements of Specification MIL-F-46370 do not form a part of this Technical Scope of Work:

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APPENDIX V (Cont'd)

a. Packaging Data (3.8)

b. Provisioning List (3.12)"

3. A separate sub-paragraph under "General Requirements" should be used to introduce minor modifications, clarifications, and supplemental requirements. Reference should be made to the specific paragraph of specification MIL-F-46370, as in the following example:

"3.1.2 Supplemental general requirements

"a. Production rates. - With regard to Paragraph 3.1.2, Specification MIL-F-46370, production designs shall be compatible with anticipated contractual quantities of 200 to 500 units, with a delivery rate of 25 to 40 per month.

"b. Human engineering. - In addition to the factors listed in paragraph 3.1.3, Specification MIL-F-46370, consideration shall be given to ease of ingress and egress through the Commander's Cupola.

"c. Environmental conditions. - With regard to Paragraph 3.1.9.2a, Specification MIL-F-46370, operation below -25°F may be achieved if necessary by providing a removable baffle over the air intake."

4. Quality Assurance Provisions

a. It is recommended that the first Quality Assurance paragraph of the Technical Scope of Work include the essence of the following:

"4.1 General quality assurance provisions. - The provisions of Paragraph 4, Specification MIL-F-46370, in their entirety, form a part of the quality assurance provisions of this Technical Scope of Work."

b. If the contract will not include the fabrication of prototype models, the following example is applicable:

"4. General quality assurance provisions. - The provisions of Paragraph 4, Specification MIL-F-46370, except for Paragraphs 4.1.1.2 and 4.3, form a part of this Technical Scope of Work."

5. Packaging, Packing and Marking Requirements

a. The packaging, packing and marking requirements of this specification may be introduced into the Technical Scope of Work as follows:

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APPENDIX V (Cont'd)

"5. Preparation for delivery. - The provisions of Section 5, Specification MIL-F-46370(MU) in their entirety, form a part of this Technical Scope of Work."

b. If the contractor will not be responsible for the development of Packaging Data (Paragraph 3.8, MIL-F-46370), the Technical Scope of Work will include specific packaging, packing and marking requirements for the prototypes and spare parts.

C. Specific Ordering Data

The following requirements will be considered in the implementation of Specification MIL-F-46370:

1. Production rates upon which product engineering should be based. (3.1.2)

2. Any proprietary features of developmental model for which non-proprietary substitution is desired. (3.1.7)

3. Any special environmental requirements or exceptions to environmental requirements of paragraph 3.1.9.

4. Items which are to be specifically covered by military specifications (without excluding additional specifications which may be required by 3.4).

5. Classes of test equipment to be designed if other than as specified in 3.5.

6. Determination of field and depot test equipment and tools to be designed, if Paragraph 3.7 is applicable.

7. Requirement for re-usable containers (3.8.1).

8. Requirement for microfilm aperture cards (3.9.3). If required, the Technical Scope of Work will include an appendix specifying detail requirements and instructions.

9. Quantity of prototypes and delivery schedule (3.10), spare parts to be provided, and inspection equipment or Field and Depot tools and test equipment to be delivered, if any. Insure that performance requirements specified in the Technical Scope of Work are suitable for determining acceptability of prototypes.

10. Maintenance of technical data during production (3.11.3). This requirement normally excluded except under a combined engineering-production contract or a continuing design agency contract.

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APPENDIX V (Cont'd)

D. Schedules

The Technical Scope of Work will preferably include only technical requirements; delivery data will normally be included in the contract schedule. Where time-phasing of various requirements is critical to timely technical accomplishment, detailed schedules will be developed in terms of time from date of award of contract. Where it is desired that the contractor develop, as part of his proposal, the detailed schedule in consonance with end delivery requirements, the following requirement may be included in the Technical Scope of Work:

> "3.1.3 Schedule for submission of technical data. - Design studies, drawings, specifications, lists and other technical data shall be submitted in accordance with time-phased schedules detailed by the contractor in his formal proposal and accepted by the Government as compatible with specified contract delivery schedules for prototypes and the final technical data package."

E. Contractor Data Requirements List

The following constitutes a list of all data of this specification for use in compiling DD Form 1423, Contractor Requirements List:

DATA TITLE	MIL-F-46370 PARA. NO.	ARMY AUTHORIZATION CODE (AMCC 715-12)
1. Specifications	3.4,3.5.4.2	04-001-0800
2. Engineering Directives and Records	3.9, 3.11	05-001-0700
3. Technical Reports	3.13	05-005-0200
4. Cards, EDMS	3.9.3	05-008-1500
5. Reproductions	3.9.2, 3.13.3	05-011-1800
6. Drawings, Engineerings, and Associated Lists	3.2, 3.6, 3.7.4	05-016-0200
7. Report of Drawing Numbers Assigned	3.2.5	05-019-0700
8. Value Engineering Data Report	3.1.4.2	05-020-0700
9. Product Engineering Studies	3.1.10	05-039-0700

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DATA TITLE	MIL-F-46370 PARA. NO.	ARMY AUTHORIZATION CODE (AMCC 715-12)
10. Quality Assurance Plan	4.1.1.1, 4.1.1.2	07-001-3900
11. Supplementary Quality Assurance Provisions	3.3	07-002-1700
12. Drawings, Inspection Equipment	3.5.4	07-004-1800
13. Contractor Certification of Compliance	4.3.4, 4.2.1.2	07-015-0700
14. Provisioning Requirements for Army Equip.	3.12	11-001-1500
15. Preservation and Packaging Data	3.8	11-024-0800

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